

Index: 1.0

Use Case: Getting Started with the FortiGate Firewall

Objective Title: Introduction

Points: 0

----- Objective Section -----

Objective Text:



## Getting Started with the FortiGate Firewall Lab

Today's networks are highly complex environments with borders that are constantly changing. In response to this highly complex environment, firewalls have become robust multi-functional devices that counter an array of threats to your network.

In this workshop, participants learn the basics of how to install a FortiGate and use it to protect a network.

FortiGate enables security-driven networking and consolidate industry-leading security capabilities, such as SSL inspection, antivirus, web filtering, and application control. By doing this, FortiGate meets the performance needs of highly scalable, hybrid IT architectures, enabling organizations to reduce complexity and manage security risks.

FortiGate simplifies security complexity and provide visibility into applications, users, and networks. FortiGate utilizes purpose-built security processing units (SPUs) and threat intelligence services from FortiGuard Labs to deliver top-rated security and high performance threat protection.

**Index:** 1.0 (a)

**Use Case:** Getting Started with the FortiGate Firewall

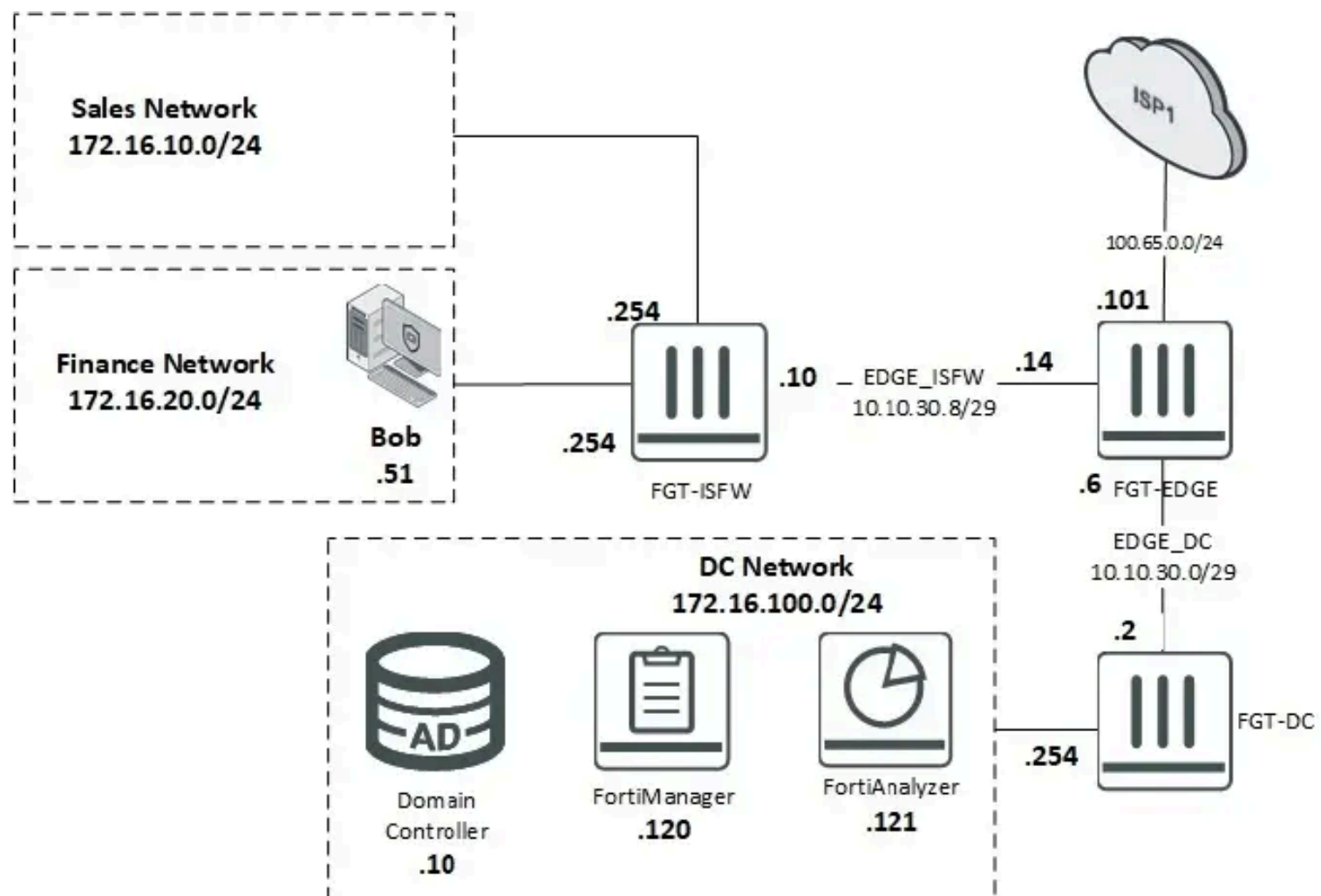
**Objective Title:** Topology

**Points:** 0

----- **Objective Section** -----

**Objective Text:**

## Topology



**Index:** 1.0 (b)  
**Use Case:** Getting Started with the FortiGate Firewall  
**Objective Title:** Agenda  
**Points:** 0

----- **Objective Section** -----  
**Objective Text:**

Agenda

This lab includes the following topics.

|          | Topic                                 | Time       |
|----------|---------------------------------------|------------|
| Lab 2.0: | FortiGate Installation                | 20 Minutes |
| Lab 3.0: | Basic Routing and Firewall Policies   | 15 Minutes |
| Lab 4.0: | Security Profiles                     | 20 Minutes |
| Lab 5.0: | Logging and Reporting                 | 10 Minutes |
| Lab 6.0: | CLI Basics                            | 15 Minutes |
| Lab 7.0: | Local User Authentication             | 20 Minutes |
| Lab 8.0: | Setting up a Fortinet Security Fabric | 10 Minutes |

**Index:** 1.0 (c)

**Use Case:** Getting Started with the FortiGate Firewall

**Objective Title:** Instructions

**Points:** 0

----- **Objective Section** -----

**Objective Text:**

## Instructions

Some of the lab exercises have a **Solve** button located at the bottom of the screen. If you click **Solve**, a script runs that completes the exercise for you. If the exercise has a Stop and Think question that includes hints, all hints will be shown when you click **Solve**. After the script runs, you must click **Continue** to move on to the next exercise.

Unless otherwise indicated all username/passwords for the various web consoles are:

- Username: `admin`
- Password: `Fortinet1!`

**Index:** 2.0

**Use Case:** FortiGate Installation

**Objective Title:** New Objective

**Points:** 0

----- **Objective Section** -----

**Objective Text:**

## FortiGate Installation

In this lab, you install a FortiGate, called FGT-EDGE, between the internet and AcmeCorp's network.

The steps you complete to accomplish this lab are:

1. Connect to the FortiGate GUI
2. Add a default route
3. Select DNS servers
4. Set the system time
5. Manage administrator accounts
6. Configure private interfaces

Time to complete: 20 minutes

## Connect to the FortiGate GUI

### Background

In this exercise, you connect to the FortiGate GUI and explore the pre-configured Management interface.

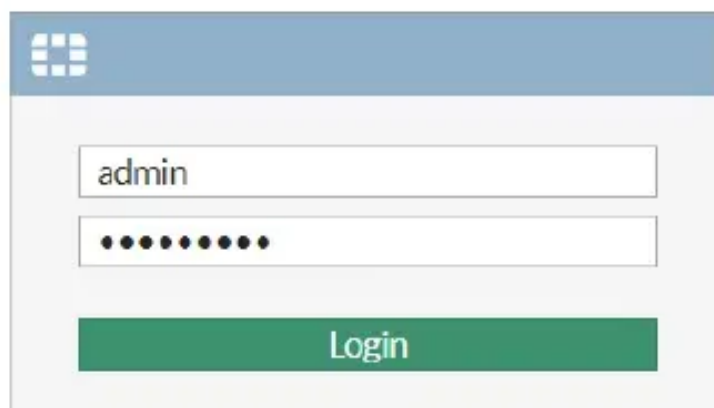
Port1 on FGT-EDGE has been pre-configured to include the following settings, which are not part of the default FortiGate configuration:

- **IP/Netmask:** 192.168.0.101/255.255.255.0
- **Administrative Access:** HTTPS, HTTP, PING, FMG-Access, SSH, and Security Fabric Connection

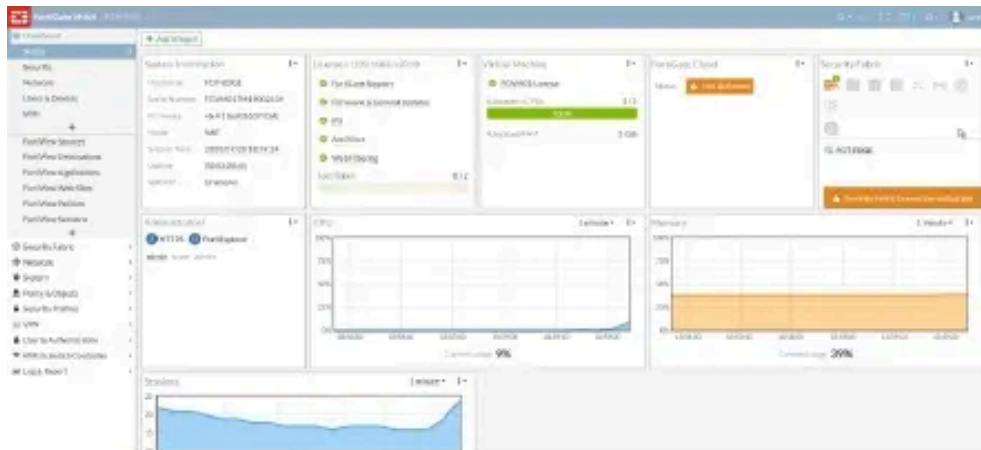
A password was also set for the default **admin** account.

### Tasks

1. Return to the Lab Activity Tab. Click **FGT-EDGE** in the sidebar menu under the Core group, and then click on the **HTTPS** option to access the **FGT-EDGE** device.
2. Log in using the default admin account by entering the following credentials:  
**Username:** admin  
**Password:** Fortinet1!



3. You have access to the FortiGate GUI.



- Click **Network > Interfaces** and select **Management Network (port1)**. Click **Edit**. You can also double-click the interface.

**Note:** Don't change any of the settings currently configured for port 1.

| <a href="#">+ Create New</a>      | <a href="#">Edit</a>      | <a href="#">Delete</a> | <input type="text" value="Search"/> | <a href="#">Q</a> |
|-----------------------------------|---------------------------|------------------------|-------------------------------------|-------------------|
| Name                              | Type                      | Members                | IP/Netmask                          |                   |
| <b>Physical Interface</b>         |                           |                        |                                     |                   |
| Dirty Network (port5)             | Physical Interface        |                        | 10.10.30.22/255.255.255.248         |                   |
| DMZ Network (port2)               | Physical Interface        |                        | 10.10.30.50/255.255.255.240         |                   |
| EDGE_DC Network (port3)           | Physical Interface        |                        | 10.10.30.6/255.255.255.248          |                   |
| EDGE_ISFW Network (port4)         | Physical Interface        |                        | 10.10.30.14/255.255.255.248         |                   |
| ISP1 (port6)                      | Physical Interface        |                        | 100.64.1.101/255.255.255.0          |                   |
| ISP2 (port7)                      | Physical Interface        |                        | 100.65.2.101/255.255.255.0          |                   |
| <b>Management Network (port1)</b> | <b>Physical Interface</b> |                        | <b>192.168.0.101/255.255.255.0</b>  |                   |

- The pre-configured settings appear under **Address** and **Administrative Access**.

Name   
 Alias   
 Type   
 Role

**Address**  
 Addressing mode: ☒ Manual ☐ DHCP ☐ Auto-managed by FortiPAM ☐ One-Arm Sniffer  
 IP/Netmask:   
 Secondary IP address: ☐

**Administrative access**  
 IPv4: ☒ HTTPS ☒ HTTP ☒ PING ☐ SNMP  
☒ FMG-Access ☒ SSH ☐ RADIUS Accounting ☒ Security Fabric Connection  
☐ FTM ☐ RADIUS Accounting

Receive LLDP: ☒ Use VDOM Setting ☐ Enable ☐ Disable  
 Transmit LLDP: ☒ Use VDOM Setting ☐ Enable ☐ Disable

- Click **Cancel** to exit without changing any settings.

## Stop and Think

Security best practices recommend configuring management interfaces with the minimal level of administrative access required. The level of access is usually based on; the role of the interface, accessibility to the interface, and the level of authority for users with access to that interface.

Consider an organization which has the following infrastructure deployed:

- FortiGate management using FortiManager Cloud services
- FortiGate two-factor authentication via FortiToken Mobile
- Remote APs participating in the organization's Security Fabric

Which of the following **Administrative Access** options should be enabled to meet the requirements for the minimal level of access for the WAN interface? (Select all that apply)

---

### Hint 1 Section

Hint: 1      Points: 2

Hint Text:

## Hint 1

Security best practices suggest only the minimum necessary administrative access be made available for an interface based on the role of the interface, accessibility to the interface, and the level of authority for users capable of accessing that interface. For a WAN interface, which of the following Administrative Access options in the FortiGate would *not* be considered minimum necessary access?

Several of the available administrative access options are Fortinet specific:

- **HTTP, HTTPS, and SSH:** These administrative access controls are meant only for administering the FortiGate firewall. They are not used for any other access such as SSL-VPN. It is not recommended to use these access controls for an interface that is directly accessible from outside your network such as the Internet.
- **FMG-Access:** Allows FortiManager, FortiManager Cloud, and FortiGate Cloud to communicate with the FortiGate for central management. It's recommended that FortiManager access always be over private networks such as VPN or MPLS, however, with the use of Fortinet Cloud services, it would be required to be available from the Internet and would be considered minimum administrative access.
- **FTM:** Allows FortiToken Mobile to use push notifications to end users as part of two-factor authentication. For remote users, this access would be required minimum administrative

access.

- **Security Fabric Connection:** Allows communication between different devices that are part of the Fortinet Security Fabric using either the FortiTelemetry or CAPWAP protocols. Devices such as FortiGate, FortiAP, FortiSwitch, FortiAnalyzer, and FortiClient use these protocols to communicate. Similar to FortiManager access, connectivity over the Internet may be required if private connections are not available. In the use case of a Remote AP, CAPWAP access from the Internet would be considered a minimum administrative access requirement.

---

#### Answer Section

**Answer:** checkbox

**Answer Text:**

### Answer

Correct: B & C

**HTTP, HTTPS, and SSH** are not considered minimum necessary administrative access. These protocols should only be enabled on trusted or private interfaces.

**FMG-Access** uses the proprietary FortiGate to FortiManager Protocol (FGFM) and would be required for FortiGate firewalls managed by FortiManager Cloud or FortiGate Cloud services.

Enabling **FTM** allows users outside the network to receive a push notification to the FortiToken Mobile app as part of the two-factor authentication process and would be considered a minimum necessary administrative access where this feature is being used.

**Security Fabric Connection** includes both the FortiTelemetry and CAPWAP protocols. CAPWAP would be required for remote APs where a VPN connection cannot be established first such as when installed at a home office and would be considered a minimum necessary administrative access control in this type of configuration.

**Answer Key:**

- ✗ 1. HTTP, HTTPS, and SSH
- ✓ 2. FMG-Access
- ✓ 3. FTM and Security Fabric Connection

**Index:** 2.0 (b)

**Use Case:** FortiGate Installation

**Objective Title:** Add a Default Route

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

## Add a Default Route

### Background

In this exercise, you add a default route to the FortiGate that the FortiGate uses to send traffic outside of the internal network.

### Tasks

1. Click **Network > Static Routes** and click **Create New**.
2. Set **Destination** to **Subnet** and leave the destination IP address set to 0.0.0.0/0.0.0.0.
3. Set **Gateway Address** to 100.65.0.254, which is the IP address you use to reach AcmeCorp's ISP.
4. Set **Interface** to **ISP1 (port6)**, the internet-facing interface.

|                           |                                                               |
|---------------------------|---------------------------------------------------------------|
| Destination ⓘ             | <div>Subnet Internet Service</div> <div>0.0.0.0/0.0.0.0</div> |
| Gateway Address           | <div>100.65.0.254</div>                                       |
| Interface                 | <div>ISP1 (port6) ▼</div>                                     |
| Administrative Distance ⓘ | <div>10</div>                                                 |
| Comments                  | <div>Write a comment... 0/255</div>                           |
| Status                    | <div>Enabled Disabled</div>                                   |

5. Click **OK**.
6. To test internet connectivity, click >\_ in the top right-hand corner to connect to the CLI console.

7. Type the command `execute ping 8.8.8.8` and press Enter.

8. The FortiGate connects to the internet, producing an output similar to the screenshot below:

```
FGT-EDGE # execute ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8): 56 data bytes
64 bytes from 8.8.8.8: icmp_seq=0 ttl=118 time=1.9 ms
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=16.1 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=13.2 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=15.8 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=13.5 ms

--- 8.8.8.8 ping statistics ---
5 packets transmitted, 5 packets received, 0% packet loss
round-trip min/avg/max = 1.9/12.1/16.1 ms
```

9. Close the CLI console by clicking on the X in the upper right corner.

**Index:** 2.0 (c)  
**Use Case:** FortiGate Installation  
**Objective Title:** Select DNS Servers  
**Points:** 10

----- **Objective Section** -----  
**Objective Text:**

Select DNS Servers

Background

In this exercise, you configure FGT-EDGE to use a FortiGuard DNS server as the primary server and the public Google DNS server as the secondary server.

**Note:** The default DNS settings use FortiGuard DNS servers as the primary and secondary servers, which may be sufficient for networks that don't have an internal DNS server.

Tasks

- 1. Click **Network > DNS**.
- 2. Set **DNS Servers** to **Specify**.
- 3. Leave **Primary DNS Server** as the default server.
- 4. Set **Secondary DNS Server** to 8 . 8 . 8 . 8 , the IP address of the Google DNS server.

|                      |                                |
|----------------------|--------------------------------|
| DNS Servers          | Use FortiGuard Servers Specify |
| Primary DNS Server   | 208.91.112.53                  |
| Secondary DNS Server | 8.8.8.8                        |
| Local Domain Name    |                                |
|                      |                                |
| DNS over TLS ⓘ       | Disable Enable Enforce         |

- 5. Select **Apply**.

Stop and Think

If the AcmeCorp network had an internal DNS server, would you configure it as the primary server or the secondary?

----- **Hint 1 Section** -----

**Hint: 1**

**Points: 2**

**Hint Text:**

## Hint

A FortiGate sends DNS requests to the primary server first, with the secondary server used as a backup.

## ----- Answer Section -----

**Answer:** radio

**Answer Text:**

## Answer

Correct: A

To make sure that the FortiGate sends DNS requests to the appropriate server, an internal server should be set as the primary DNS server.

**Answer Key:**

- ✓ 1. Primary
- ✗ 2. Secondary

**Index:** 2.0 (d)

**Use Case:** FortiGate Installation

**Objective Title:** Set the System Time

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

## Set the System Time

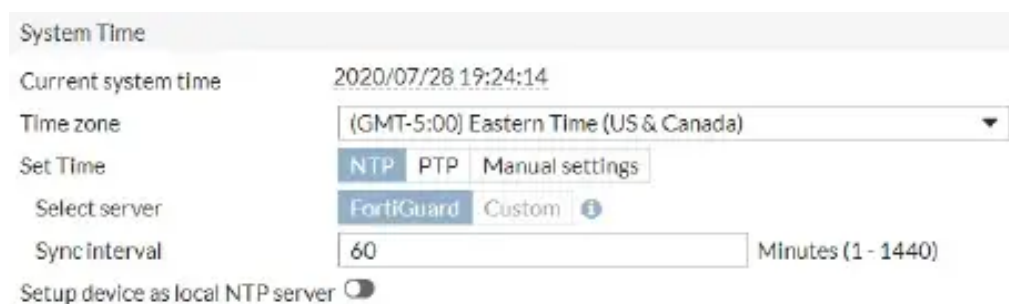
### Background

In this exercise, you configure the system time on FGT-EDGE to AcmeCorp's local time zone, Eastern Standard Time.

**Note:** For the purpose of this lab, you must select Eastern Standard Time. Making changes to the time zone could disrupt the lab functionality.

### Tasks

1. Click **System > Settings**.
2. Under **System Time**, select **(GMT-5:00) Eastern Time (US & Canada)**.
3. Set **Set Time** to **NTP**.
4. Set **Select server** to **FortiGuard**.



The screenshot shows the 'System Time' configuration page in a web interface. It includes the following fields and options:

- Current system time:** 2020/07/28 19:24:14
- Time zone:** A dropdown menu showing '(GMT-5:00) Eastern Time (US & Canada)'.
- Set Time:** Three tabs: 'NTP' (selected), 'PTP', and 'Manual settings'.
- Select server:** Two tabs: 'FortiGuard' (selected) and 'Custom' with an information icon.
- Sync interval:** A text input field containing '60', with a label 'Minutes (1 - 1440)'.
- Setup device as local NTP server:** A toggle switch that is currently turned off.

5. Select **Apply**.

**Index:** 2.0 (e)

**Use Case:** FortiGate Installation

**Objective Title:** Configure Private Interfaces

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

## Configure Private Interfaces

### Background

In AcmeCorp's network, there are two additional FortiGate devices: FGT-ISFW and FGT-DC. These devices connect to FGT-EDGE to reach the internet.

In this exercise, you configure two private interfaces on FGT-EDGE: port 4 that connects to FGT-ISFW and port 3 that connects to FGT-DC.

### Tasks

1. Click **Network > Interfaces**, select **port4**, and then click **Edit**.
2. Configure the following settings:
  - **Alias:** EDGE\_ISFW Network
  - **Role:** LAN
  - **IP/Netmask:** 10.10.30.14/255.255.255.248
  - **Administrative access:** HTTPS, HTTP, PING, and Security Fabric Connection

**Note:** Enabling HTTPS access automatically enables HTTP access.

The screenshot shows the configuration page for the interface 'port4'. The 'Name' field is 'port4'. The 'Alias' field is 'EDGE\_ISFW Network'. The 'Type' is 'Physical Interface'. The 'Role' is set to 'LAN'. Under the 'Address' section, the 'Addressing mode' is 'Manual', and the 'IP/Netmask' is '10.10.30.14/255.255.255.248'. The 'Create address object matching subnet' and 'Secondary IP address' options are disabled. Under the 'Administrative access' section, for 'IPv4', the following are checked: 'HTTPS', 'HTTP', 'PING', and 'Security Fabric Connection'. 'FMG-Access', 'SSH', 'FTM', and 'RADIUS Accounting' are unchecked. At the bottom, 'Receive LLDP' and 'Transmit LLDP' are both set to 'Use VDOM Setting' with 'Enable' selected.

|                                       |                                                                                                                                        |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| Name                                  | port4                                                                                                                                  |
| Alias                                 | EDGE_ISFW Network                                                                                                                      |
| Type                                  | Physical Interface                                                                                                                     |
| Role                                  | LAN                                                                                                                                    |
| Address                               |                                                                                                                                        |
| Addressing mode                       | Manual   DHCP   Auto-managed by FortiIPAM                                                                                              |
| IP/Netmask                            | 10.10.30.14/255.255.255.248                                                                                                            |
| Create address object matching subnet | <input type="checkbox"/>                                                                                                               |
| Secondary IP address                  | <input type="checkbox"/>                                                                                                               |
| Administrative access                 |                                                                                                                                        |
| IPv4                                  | <input checked="" type="checkbox"/> HTTPS <input checked="" type="checkbox"/> HTTP <input checked="" type="checkbox"/> PING            |
|                                       | <input type="checkbox"/> FMG-Access <input type="checkbox"/> SSH <input type="checkbox"/> SNMP                                         |
|                                       | <input type="checkbox"/> FTM <input type="checkbox"/> RADIUS Accounting <input checked="" type="checkbox"/> Security Fabric Connection |
| Receive LLDP                          | Use VDOM Setting   Enable   Disable                                                                                                    |
| Transmit LLDP                         | Use VDOM Setting   Enable   Disable                                                                                                    |

3. Click **OK**.
4. Click **Network > Interfaces**, select **port3**, and select **Edit**.
5. Configure the following settings:
  - **Alias:** EDGE\_DC Network
  - **Role:** LAN
  - **IP/Netmask:** 10.10.30.6/255.255.255.248
  - **Administrative access:** PING and Security Fabric Connection

Name: port3

Alias: EDGE\_DC Network

Type: Physical Interface

Role: LAN

Addressing mode: Manual DHCP Auto-managed by FortiIPAM

IP/Netmask: 10.10.30.6/255.255.255.248

Create address object matching subnet: ☐

Secondary IP address: ☐

Administrative access:

IPv4:

- ☐ HTTPS
- ☐ FMG-Access
- ☐ FTM
- ☐ HTTP
- ☐ SSH
- ☐ RADIUS Accounting
- ☒ PING
- ☐ SNMP
- ☒ Security Fabric Connection

Receive LLDP: Use VDOM Setting Enable Disable

Transmit LLDP: Use VDOM Setting Enable Disable

6. Click **OK**.

## Stop and Think

What else do you need to configure on FGT-EDGE to allow internet access for the networks behind FGT-ISFW and FGT-DC? (Select all that apply)

----- Hint 1 Section -----

**Hint: 1**      **Points: 2**

**Hint Text:**

## Hint

FGT-ISFW and FGT-DC are already configured to receive DNS server and system time settings directly from FGT-EDGE, so you don't need to configure them separately for the other FortiGate devices.

----- **Answer Section** -----

**Answer:** checkbox

**Answer Text:**

## Answer

Correct: B and C

In order to allow internet access for the networks behind FGT-ISFW and FGT-DC, there must be static routes for these networks, as well as firewall policies to allow traffic to pass through FGT-EDGE to the internet.

You will configure routing and firewall policies in the next objective.

**Answer Key:**

- ☐ 1. Additional DNS servers
- ☒ 2. Static routes for each network
- ☒ 3. Firewall policies to allow internet access
- ☐ 4. System times for the other FortiGate devices

**Index:** 3.0

**Use Case:** Basic Routing and Firewall Policies

**Objective Title:** New Objective

**Points:** 0

----- **Objective Section** -----

**Objective Text:**

## Basic Routing and Firewall Policies

In AcmeCorp's network, FGT-ISFW has two networks behind it: Sales (172.16.10.0/24) and Finance (172.16.20.0/24). FGT-DC has one network behind it, DC (172.16.100.0/24).

In this lab, you configure basic routing and firewall policies on FGT-EDGE to allow devices on these networks to connect to the internet.

The steps you complete to accomplish this lab are:

1. Create firewall addresses and an address group
2. Create additional static routes
3. Create firewall policies

Time to complete: 15 minutes

**Index:** 3.0 (a)

**Use Case:** Basic Routing and Firewall Policies

**Objective Title:** Create Firewall Addresses and an Address Group

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

## Create Firewall Addresses and an Address Group

### Background

Firewall addresses define sources and destinations of network traffic and are used when creating firewall policies. Address groups are used to group together firewall addresses that require the same firewall policy.

In this exercise, you create three firewall addresses, one for each network. You also create a firewall group that contains the addresses for the Sales and Finance networks.

### Tasks

1. Click **Policy & Objects > Addresses** and then use the **Create New** drop-down menu to select **Address** and create an address for the Sales network.
2. Configure the following settings:
  - **Name:** Sales
  - **Type:** Subnet
  - **IP/Netmask:** 172.16.10.0/24
  - **Interface:** any
3. Turn on **Static route configuration**.

|                            |                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Name                       | <input type="text" value="Sales"/>                                                                                        |
| Color                      |  <input type="button" value="Change"/> |
| Type                       | <input type="text" value="Subnet"/>                                                                                       |
| IP/Netmask                 | <input type="text" value="172.16.10.0/24"/>                                                                               |
| Interface                  | <input type="text" value="any"/>                                                                                          |
| Static route configuration | <input checked="" type="checkbox"/>                                                                                       |
| Comments                   | <input type="text" value="Write a comment..."/> 0/255                                                                     |

4. Click **OK**.
5. Click **Create New > Address** to create an address for the Finance network.

6. Configure the following settings:

- **Name:** Finance
- **Type:** Subnet
- **IP/Netmask:** 172.16.20.0/24
- **Interface:** any

7. Turn on **Static route configuration**.

|                            |                                                                                                                         |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Name                       | <input type="text" value="Finance"/>                                                                                    |
| Color                      |  <input type="button" value="Change"/> |
| Type                       | <input type="text" value="Subnet"/>                                                                                     |
| IP/Netmask                 | <input type="text" value="172.16.20.0/24"/>                                                                             |
| Interface                  | <input type="text" value="any"/>                                                                                        |
| Static route configuration | <input checked="" type="checkbox"/>                                                                                     |
| Comments                   | <input type="text" value="Write a comment..."/> 0/255                                                                   |

8. Click **OK**.

9. Click **Create New > Address** to create an address for the DC network.

10. Configure the following settings:

- **Name:** DC
- **Type:** Subnet
- **IP/Netmask:** 172.16.100.0/24
- **Interface:** any

11. Turn on **Static route configuration**.

|                            |                                                                                                                           |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Name                       | <input type="text" value="DC"/>                                                                                           |
| Color                      |  <input type="button" value="Change"/> |
| Type                       | <input type="text" value="Subnet"/>                                                                                       |
| IP/Netmask                 | <input type="text" value="172.16.100.0/24"/>                                                                              |
| Interface                  | <input type="text" value="any"/>                                                                                          |
| Static route configuration | <input checked="" type="checkbox"/>                                                                                       |
| Comments                   | <input type="text" value="Write a comment..."/> 0/255                                                                     |

12. Click **OK**.

13. Use the **Create New** drop-down menu to click **Address Group**.

14. Configure the following settings:

- **Group name:** Sales and Finance

- **Type: Group**
- **Members: Finance and Sales**

15. Turn on **Static route configuration**.

Group name: Sales and Finance

Color: Change

Type: **Group** Folder

Members: Finance, Sales

Exclude members: ☐

Static route configuration: ☒

Comments: Write a comment... 0/255

16. Click **OK**.

## Stop and Think

By creating an address group that contains the addresses for both Sales and Finance, you can now configure FGT-EDGE to treat traffic from both of these networks in the exact same way. What reasons might there be to not use this group and instead have FGT-EDGE distinguish between traffic from Sales and traffic from Finance? (Select all that apply)

### ----- Hint 1 Section -----

**Hint: 1**      **Points: 2**

**Hint Text:**

## Hint 1

Using address groups makes it easier to configure multiple policies that all use the same addresses, since to add or remove an address from these policies, you only need to make a single edit to the group configuration.

### ----- Hint 2 Section -----

**Hint: 2**      **Points: 2**

**Hint Text:**

## Hint 2

Firewall groups do not affect the Security Fabric topology views, so the two networks will appear separately even if you use the firewall group.

### ----- Answer Section -----

**Answer:** checkbox

**Answer Text:**

## Answer

Correct: B and D

Both security and routing requirements are reasons to have FGT-EDGE handle traffic differently depending on whether it came from Sales or Finance.

**Answer Key:**

- ☐ 1. It's easier to use individual addresses in all situations
- ☒ 2. The Finance network has greater security requirements
- ☐ 3. Both networks need to appear separately in the Security Fabric topology views
- ☒ 4. You wish to use different routing settings for the two networks

**Index:** 3.0 (b)

**Use Case:** Basic Routing and Firewall Policies

**Objective Title:** Create Additional Static Routes

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

## Create Additional Static Routes

### Background

In this exercise, you create two static routes: one for Sales and Finance and one for DC.

### Tasks

1. Click **Network > Static Routes** and click **Create New** to create a static route for Sales and Finance.
2. Configure the following settings:
  - **Destination: Named Address**
  - Use the drop-down menu to select **Sales and Finance**
  - **Gateway Address:** 10.10.30.10
  - **Interface:** EDGE\_ISFW Network (port4)
  - **Comments:** Sales and Finance

New Static Route

Destination: Subnet | **Named Address** | Internet Service

Sales and Finance

Gateway Address: 10.10.30.10

Interface: EDGE\_ISFW Network (port4)

Administrative Distance: 10

Comments: Sales and Finance 17/255

Status: **Enabled** Disabled

3. Click **OK**.
4. Click **Create New** to create a static route for DC.
5. Configure the following settings:
  - **Destination: Named Address**

- Use the drop-down menu to select **DC**
- **Gateway Address:** 10.10.30.2
- **Interface:** EDGE\_DC Network (port3)
- **Comments:** DC

New Static Route

Destination: Subnet | **Named Address** | Internet Service

Gateway Address: 10.10.30.2

Interface: EDGE\_DC Network (port3)

Administrative Distance: 10

Comments: DC

Status: ☒ Enabled ☐ Disabled

6. Click **OK**.

## Stop and Think

True or false: Bob's computer, located on the Finance network, now has internet access.

----- **Answer Section** -----

**Answer:** radio

**Answer Text:**

## Answer

Correct: B

False. If you connect to Bob's computer now, you aren't able to access the internet. This is because there is no firewall policy in place to allow traffic from the Finance network to pass through FGT-EDGE.

**Answer Key:**

- ☒ 1. True
- ☐ 2. False

**Index:** 3.0 (c)  
**Use Case:** Basic Routing and Firewall Policies  
**Objective Title:** Create Firewall Policies  
**Points:** 10

----- **Objective Section** -----  
**Objective Text:**

Create Firewall Policies

Background

In this exercise, you create two firewall policies: one for Sales and Finance and one for DC.

Tasks

1. Click **Policy & Objects > Firewall Policy** and click **Create New** to create a policy for Sales and Finance.
2. Configure the following settings:
  - **Name:** Sales and Finance
  - **Incoming Interface:** EDGE\_ISFW Network (port4)
  - **Outgoing Interface:** ISP1 (port6)
  - **Source:** Sales and Finance
  - **Destination:** all
  - **Service:** ALL
3. Leave all other settings at the default.

|                    |                                                                                                                                 |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Name ⓘ             | Sales and Finance                                                                                                               |
| Incoming Interface |  EDGE_ISFW Network (port4) ▼                 |
| Outgoing Interface |  ISP1 (port6) ▼                              |
| Source             | <div> Sales and Finance ✕</div> <div>+</div> |
| Destination        | <div> all ✕</div> <div>+</div>               |
| Schedule           |  always ▼                                    |
| Service            | <div> ALL ✕</div> <div>+</div>               |
| Action             | <div><input checked="" type="checkbox"/> ACCEPT <input type="checkbox"/> DENY</div>                                             |

4. Click **OK**.

5. Click **Create New** to create a policy for DC.
6. Configure the following settings:
- **Name:** DC
  - **Incoming Interface:** EDGE\_DC Network (port3)
  - **Outgoing Interface:** ISP1 (port6)
  - **Source:** DC
  - **Destination:** all
  - **Service:** ALL
7. Leave all other settings at the default.

The screenshot shows a policy configuration form with the following fields and values:

- Name:** DC
- Incoming Interface:** EDGE\_DC Network (port3)
- Outgoing Interface:** ISP1 (port6)
- Source:** DC
- Destination:** all
- Schedule:** always
- Service:** ALL
- Action:** ACCEPT (selected), DENY
- Inspection Mode:** Flow-based (selected), Proxy-based

8. Click **OK**.
9. In the top right-hand corner of the screen, click **By Sequence**. The policy list is now displayed in the order FGT-EDGE uses to match traffic, with the default **Implicit Deny** policy at the bottom.

| + Create New      | Edit                     | Delete       | Policy Lookup     | Search      | Interface Pair View | By Sequence |
|-------------------|--------------------------|--------------|-------------------|-------------|---------------------|-------------|
| Name              | From                     | To           | Source            | Destination | Schedule            | Service     |
| Sales and Finance | EDGE_ISFWNetwork (port4) | ISP1 (port6) | Sales and Finance | all         | always              | ALL         |
| DC                | EDGE_DC Network (port3)  | ISP1 (port6) | DC                | all         | always              | ALL         |
| Implicit Deny     | any                      | any          | all               | all         | always              | ALL         |

10. Return to the Lab Activity Tab, click **Bob** in the sidebar menu under the Finance group, and then click on the **RDP** option to access Bob’s workstation.
11. Run Chrome and click the browser bookmark **Google**.

12. Bob’s computer has internet access.
13. Connect to FGT-EDGE and refresh the policy list. The **Bytes** column shows that the **Sales and Finance** policy has traffic.

Stop and Think

True or false: FGT-EDGE is now applying security scanning, such as antivirus, to traffic from the Sales, Finance, and DC network.

----- Answer Section -----

Answer: radio

Answer Text:

Answer

Correct: B

False. FGT-EDGE is processing traffic from these networks but, since no security profiles have been enabled on the firewall policy, FGT-EDGE isn’t applying security scanning.

Answer Key:

- X 1. True
- ✓ 2. False

**Index:** 4.0

**Use Case:** Security Profiles

**Objective Title:** New Objective

**Points:** 0

----- **Objective Section** -----

**Objective Text:**

## Security Profiles

In this lab, you add security profiles to the Sales and Finance firewall policy on FGT-EDGE, so that the FortiGate protects traffic from these two networks.

The steps you complete to accomplish this lab are:

1. Apply antivirus scanning and SSL inspection
2. Block social media with web filtering
3. Block Mozilla Firefox with application control

Time to complete: 20 minutes

**Index:** 4.0 (a)  
**Use Case:** Security Profiles  
**Objective Title:** Apply Antivirus Scanning and SSL Inspection  
**Points:** 10

----- **Objective Section** -----  
**Objective Text:**

## Apply Antivirus Scanning and SSL Inspection

### Background

In this exercise, you create an antivirus profile for Sales and Finance, to protect network traffic from virus outbreaks. You also apply full SSL inspection, to allow FGT-EDGE to inspect encrypted traffic.

When you apply full SSL inspection to traffic, network users may receive a security certificate warning in their internet browser. In this exercise, Bob’s computer has been pre-configured to prevent any warnings from appearing.

### Tasks

1. Return to the **FGT-EDGE** tab.
2. Click **Security Profiles > AntiVirus** and click **Create New**.
3. Set **Name** to Sales and Finance.
4. Leave **Detect Viruses** as **Block**.
5. Leave **Feature set** as **Flow-based**. Flow-based inspection takes a snapshot of content packets and uses pattern matching to identify security threats in the content. Proxy-based inspection reconstructs content that passes through the FortiGate and inspects the content for security threats.

|                |                                                                               |
|----------------|-------------------------------------------------------------------------------|
| Name           | <input type="text" value="Sales and Finance"/>                                |
| Comments       | <input type="text" value="Write a comment..."/> 0/255                         |
| Detect Viruses | <input checked="" type="radio"/> Block <input type="radio"/> Monitor          |
| Feature set    | <input checked="" type="radio"/> Flow-based <input type="radio"/> Proxy-based |

6. Under **Inspected Protocols**, turn on all protocol options.



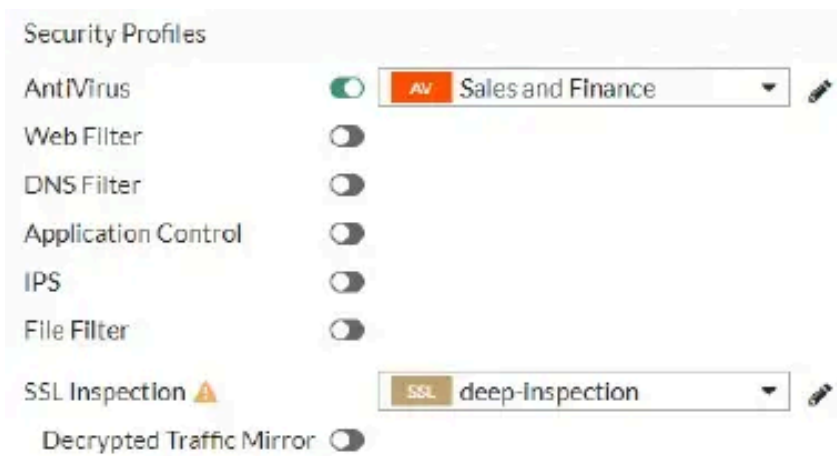
- Under **APT Protection Options**, turn on **Treat Windows Executables in Email Attachments as Viruses** and leave **Include Mobile Malware Protection** turned on.



- Under **Virus Outbreak Prevention**, turn on **Use FortiGate Outbreak Prevention Database**. This allows the FortiGate antivirus database to use third-party malware hash signatures curated by the FortiGuard to block detected viruses before a FortiGuard signature is available.



- Click **OK**.
- Click **Policy & Object > Firewall Policy**, click **Sales and Finance**, and click **Edit**.
- Under **Security Profiles**, turn on **AntiVirus**. Use the drop-down menu to select the **Sales and Finance** profile.
- Use the **SSL Inspection** drop-down menu to select **deep-inspection**. This turns on full SSL inspection, so FGT-EDGE can inspect encrypted traffic.

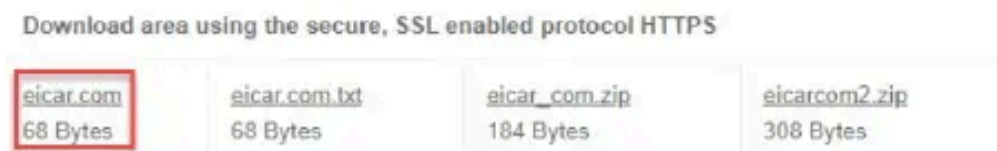


13. Click **OK**.

14. Return to **Bob's** workstation.

15. Run Chrome and click the browser bookmark **EICAR**. This website contains a file that you can use to test your antivirus scanning.

16. Under **Download area using the secure, SSL enabled protocol https**, click **eicar.com**.



17. FGT-EDGE blocks the file from downloading.



## Stop and Think

Would FGT-EDGE block the EICAR test file if you set **SSL Inspection** to **certificate-inspection** and downloaded the file using HTTPS?

----- **Hint 1 Section** -----

**Hint: 1**

**Points: 2**

**Hint Text:**

## Hint

When you apply the **certificate-inspection** profile, the FortiGate only inspects the header information of the packets and not the contents.

You can apply the following security features when using SSL certificate inspection mode: web filtering and application control. With web filtering, SSL certificate inspection doesn't introduce certificate errors and can be a useful alternative to full SSL inspection. With application control, SSL certificate inspection can use the common name in the server certificate to identify an application by certain signatures; however, most signatures require full SSL inspection.

## ----- Answer Section -----

**Answer:** radio

**Answer Text:**

## Answer

Correct: B

No, in order to inspect encrypted traffic, you must use full SSL inspection.

**Answer Key:**

☒ 1. Yes

☐ 2. No

**Index:** 4.0 (b)  
**Use Case:** Security Profiles  
**Objective Title:** Block Social Networking with Web Filtering  
**Points:** 10

----- **Objective Section** -----  
**Objective Text:**

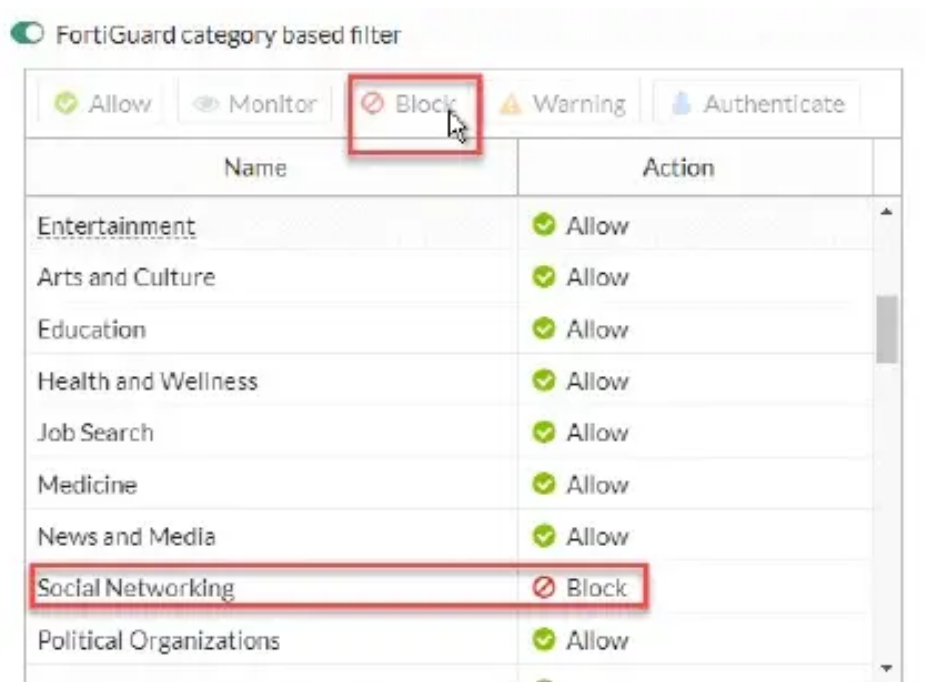
## Block Social Networking with Web Filtering

### Background

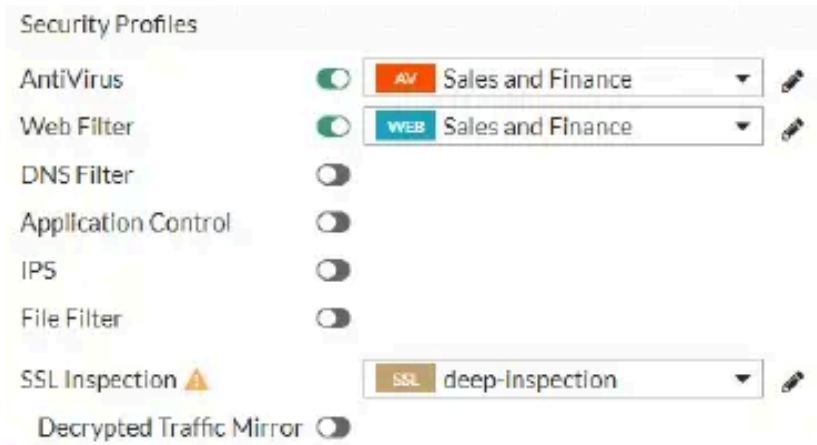
In this exercise, you block all websites identified as social networking sites by the FortiGuard Web Filtering service.

### Tasks

- 1. Return to the **FGT-EDGE**.
- 2. Click **Security Profiles > Web Filter** and click **Create New**.
- 3. Set **Name** to Sales and Finance.
- 4. Leave **Feature set** set to **Flow-based**.
- 5. Under **FortiGuard category based filter**, locate **General Interest - Personal**.
- 6. Select **Social Networking** and click **Block**.



7. Click **OK**.
8. Click **Policy & Object > Firewall Policy**, click **Sales and Finance**, and click **Edit**.
9. Under **Security Profiles**, turn on **Web Filter**. Use the drop-down menu to select the **Sales and Finance** profile.



10. Click **OK**.
11. Return to **Bob's Computer** and click the browser bookmark **Twitter**.
12. FGT-EDGE blocks the website.



## Stop and Think

If you wanted to allow access to Twitter while blocking other social network sites, what methods could you use? (Select all that apply)

----- Hint 1 Section -----

**Hint: 1**

**Points: 2**

**Hint Text:**

## Hint

You can use a static URL filter to block, allow, or monitor URLs by using patterns containing text, regular expressions, or wildcard characters.

By using a web rating override, you can manually assign a specific website to a different Fortinet category or a locally created category.

----- Answer Section -----

**Answer:** checkbox

**Answer Text:**

## Answer

Correct: A, B, and C

You can use any of these three methods to allow access to Twitter while blocking other social networking websites.

**Answer Key:**

- ✓ 1. Configure a static URL filter for Twitter
- ✓ 2. Using a web rating override to assign Twitter to a different category
- ✓ 3. Create another firewall policy to handle traffic to Twitter
- ✗ 4. There is no method to allow this

**Index:** 4.0 (c)  
**Use Case:** Security Profiles  
**Objective Title:** Block Mozilla Firefox with Application Control  
**Points:** 10

----- **Objective Section** -----  
**Objective Text:**

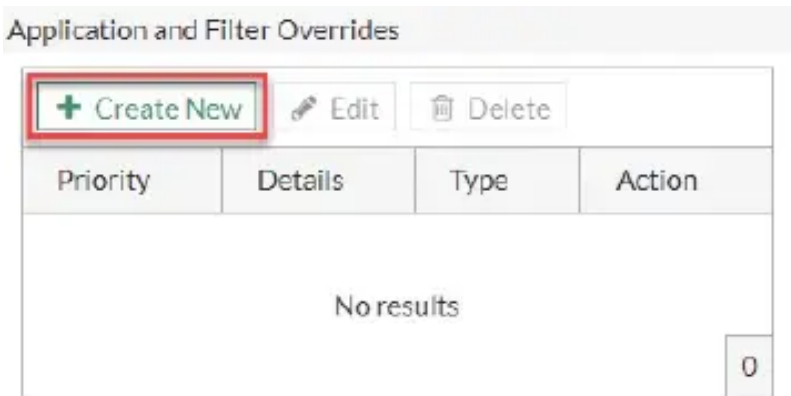
## Block Mozilla Firefox with Application Control

### Background

In this exercise, you block network users from using the browser Mozilla Firefox.

### Tasks

- 1. Return to the **FGT-EDGE**.
- 2. Click **Security Profiles > Application Control** and click **Create New**.
- 3. Set **Name** to Sales and Finance.
- 4. Under **Application and Filter Overrides**, click **Create New**.



- 5. Leave **Type** set to **Application** and **Action** set to **Block**.
- 6. In the search bar, type `Firefox` and then press `Enter`.
- 7. Two signatures are found: **Firefox.Update** and **HTTP.BROWSER\_Firefox**. Click **Add All Results**.

|                                                                                                                                                                                                                                                                                                                            |                                                                                                        |                                                                                              |               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------|
| Type                                                                                                                                                                                                                                                                                                                       | Application                                                                                            | Filter                                                                                       |               |
| Action                                                                                                                                                                                                                                                                                                                     |  Block                |                                                                                              |               |
| <div><div> Add All Results</div><div>Firefox</div><div></div></div> |                                                                                                        |                                                                                              |               |
|                                                                                                                                                                                                                                           | Name                                                                                                   | Category                                                                                     | Technology    |
| Application Signature 2/1762                                                                                                                                                                                                                                                                                               |                                                                                                        |                                                                                              |               |
|                                                                                                                                                                                                                                           |  Firefox.Update       |  Update     | Client-Server |
|                                                                                                                                                                                                                                           |  HTTP.BROWSER_Firefox |  Web.Client | Browser-Based |

8. Click **OK**.
9. Click **OK** to save the profile.
10. Click **Policy & Object > Firewall Policy**, click **Sales and Finance**, and click **Edit**.
11. Under **Security Profiles**, turn on **Application Control**. Use the drop-down menu to select **Sales and Finance**.

|                          |                                     |     |                   |
|--------------------------|-------------------------------------|-----|-------------------|
| Security Profiles        |                                     |     |                   |
| AntiVirus                | <input checked="" type="checkbox"/> | AV  | Sales and Finance |
| Web Filter               | <input checked="" type="checkbox"/> | WEB | Sales and Finance |
| DNS Filter               | <input type="checkbox"/>            |     |                   |
| Application Control      | <input checked="" type="checkbox"/> | APP | Sales and Finance |
| IPS                      | <input type="checkbox"/>            |     |                   |
| File Filter              | <input type="checkbox"/>            |     |                   |
| SSL Inspection           | <input type="checkbox"/>            | SSL | deep-Inspection   |
| Decrypted Traffic Mirror | <input type="checkbox"/>            |     |                   |

12. Click **OK**.
13. Return to **Bob's Computer** and run Firefox.
14. Click the browser bookmark **Google**.
15. FGT-EDGE blocks the attempt.



## FortiGate Application Control

### Application Blocked

You have attempted to use an application that violates your Internet usage policy.

|             |                                      |
|-------------|--------------------------------------|
| Application | HTTP.BROWSER_Firefox                 |
| Category    | Web.Client                           |
| URL         | http://google.com/                   |
| Username    |                                      |
| Group Name  |                                      |
| Policy      | 78d65486-0fd6-51eb-a091-e17420f47c67 |

### Stop and Think

Can you use the security profiles you applied to the Sales and Finance firewall policy for the DC firewall policy as well?

#### ----- Answer Section -----

**Answer:** radio

**Answer Text:**

### Answer

Correct: A

Yes, you can apply the same security profiles to all firewall policies on FGT-EDGE.

**Answer Key:**

- ✓ 1. Yes
- X 2. No

**Index:** 4.0 (d)

**Use Case:** Security Profiles

**Objective Title:** Learn More About Security Profiles

**Points:** 0

----- **Objective Section** -----

**Objective Text:**

## Learn More About Security Profiles

If you would like to learn more about using FortiGate security profiles, ask your instructor about the *Fortifying the Enterprise Network (NGFW Solution)* course offered by the Fast Track Program.

As security architects consider how to provide comprehensive threat protection for their enterprises, including intrusion prevention, web filtering, anti-malware, and application control, they face a major complexity hurdle managing these point products with no integration and lack of visibility.

Participants who attend this workshop will learn how to:

- Reduce complexity with industry-leading security effectiveness
- Enhance visibility with automated action
- Simplify SSL performance and complexity issues for encrypted cloud access

**Index:** 5.0

**Use Case:** Logging and Reporting

**Objective Title:** New Objective

**Points:** 0

----- **Objective Section** -----

**Objective Text:**

## Logging and Reporting

In this lab, you configure FortiGate local logging and view the logs generated from network traffic.

The steps you complete to accomplish this lab are:

1. Configure log settings
2. Enable logging in the firewall policy
3. Generate traffic and view the logs

Time to complete: 10 minutes

**Index:** 5.0 (a)  
**Use Case:** Logging and Reporting  
**Objective Title:** Configure Log Settings  
**Points:** 10

----- **Objective Section** -----  
**Objective Text:**

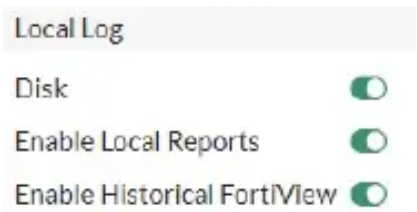
## Configure Log Settings

### Background

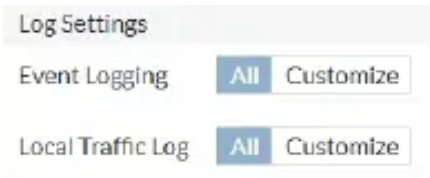
In this exercise, you configure log settings for local logging.

### Tasks

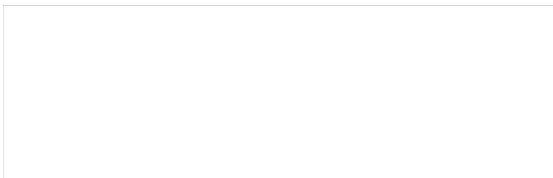
1. Return to the **FGT-EDGE**.
2. Click **Log & Report > Log Settings**.
3. Under **Local Log**, turn on **Disk**, **Enable Local Reports**, and **Enable Historical FortiView**.



4. Under **Log Settings**, set **Event Logging** and **Local Traffic Log** to **All**.



5. Under **GUI Preferences**, turn on both **Resolve Hostnames** and **Resolve Unknown Applications**.



6. Click **Apply**.

**Index:** 5.0 (b)

**Use Case:** Logging and Reporting

**Objective Title:** Enable Logging in the Firewall Policy

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

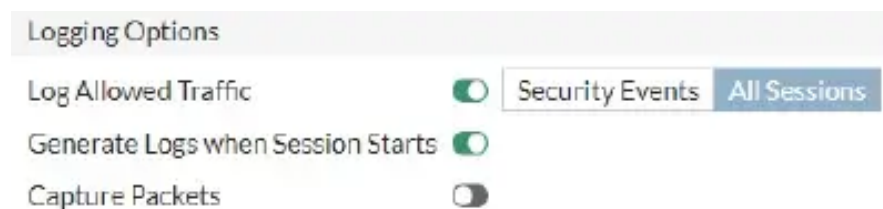
## Enable Logging in the Firewall Policy

### Background

Now that you configured local logging, you need to enable logging on a firewall policy to begin to generate logs. Because logging all sessions uses more system resources, it is typically recommended to log only security events. However, for this exercise, you configure the FortiGate to log all sessions to make sure logging is working properly.

### Tasks

1. Click **Policy & Object > Firewall Policy**, click **Sales and Finance**, and click **Edit**.
2. Under **Logging Options**, turn on **Log Allowed Traffic** and select **All Sessions**.
3. Turn on **Generate Logs when Session Starts**.



4. Click **OK**.

**Index:** 5.0 (c)

**Use Case:** Logging and Reporting

**Objective Title:** Generate Traffic and View the Logs

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

## Generate Traffic and View the Logs

### Background

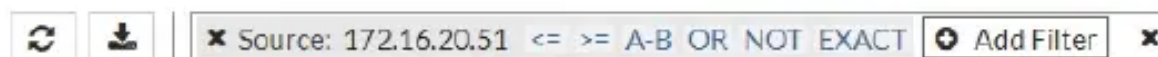
There are many types of logs you can view on a FortiGate. In this exercise, you will use the following logs:

- Forward traffic: Logs about all network traffic permitted or denied by a firewall policy
- Events: Logs concerning administration management and FortiGate system activity
- Web filter: Logs generated when web filtering is applied to network traffic

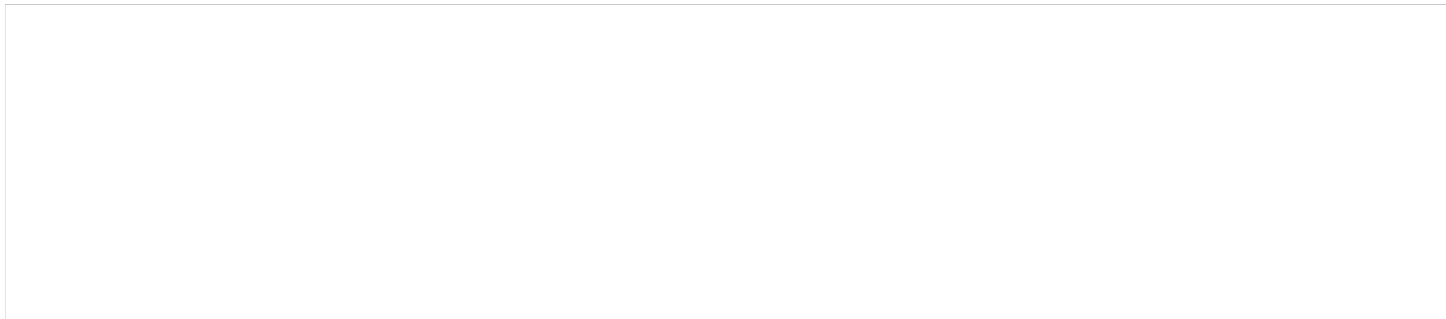
### Tasks

#### Forward Traffic

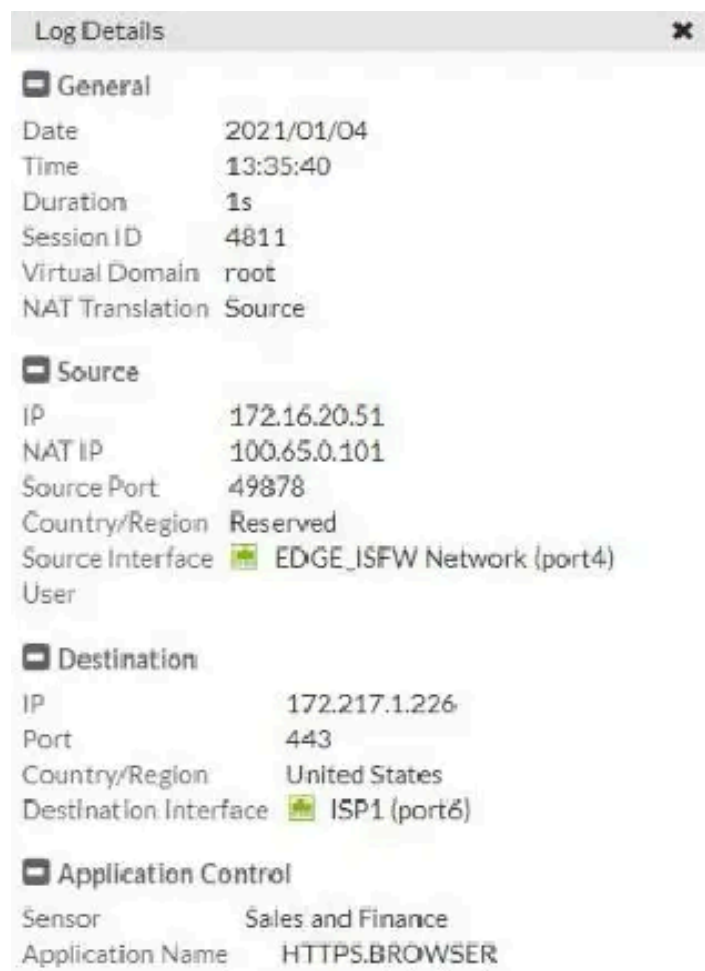
1. Return to **Bob's Computer**.
2. Run Chrome and click the browser bookmark **Google** to generate internet traffic through FGT-EDGE.
3. Return to the **FGT-EDGE**.
4. Click **Log & Report > Forward Traffic**.
5. On the top of the screen, click **Add Filter**. Select **Source** and then select **172.16.20.51**, the IP address of Bob's computer.



6. With the filter applied, the log shows the traffic from Bob's computer. The traffic should look similar to the following screenshot:



7. Select one of the log entries and click **Details**, located in the top right-hand corner. The FortiGate displays more information about the session.



## Events

1. Log out of **FGT-EDGE**.
2. Attempt to log in with the **admin** account but do *not* enter a password.
3. The attempt fails.

**Note:** Only enter the wrong credentials once, to avoid FGT-EDGE locking the **admin** account out after too many failed login attempts.

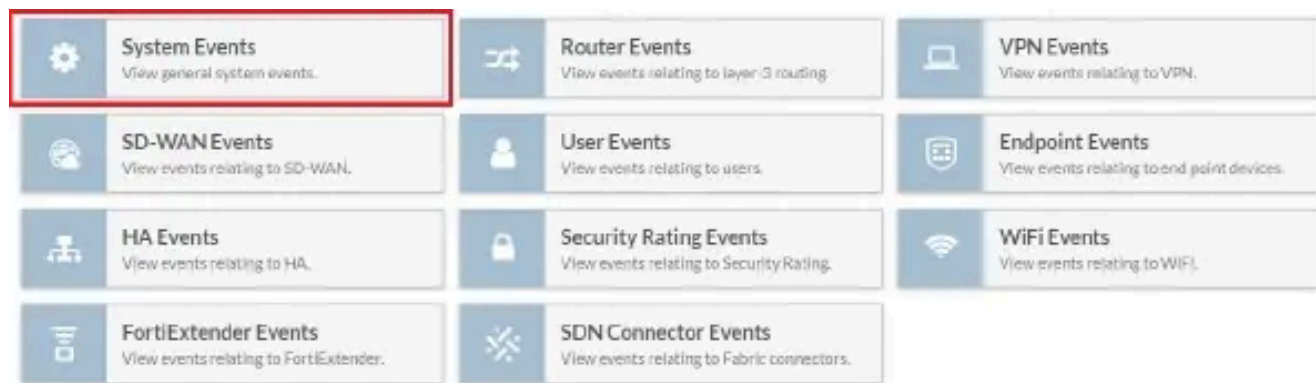
Authentication failure

Username

Password

Login

4. Log in using the correct credentials for the **admin** account (username: admin, password: Fortinet1!).
5. Click **Log & Report > Events**. Click the tile for **System Events**.



6. The event log contains an entry for the failed login attempt.

| Date/Time           | Level | User  | Message                                        | Log Description         |
|---------------------|-------|-------|------------------------------------------------|-------------------------|
| 2021/01/04 13:37:40 |       | admin | Administrator admin logged in successfully ... | Admin login successful  |
| 2021/01/04 13:37:15 |       | admin | Administrator admin login failed from https... | Admin login failed      |
| 2021/01/04 13:37:04 |       | admin | Configuration is changed in the admin sessl... | Configuration changed   |
| 2021/01/04 13:37:04 |       | admin | Administrator admin logged out from https...   | Admin logout successful |

7. Select the log entry and click **Details** to view more information.



## Web Filter

1. Return to **Bob's Computer** and click the browser bookmark **Twitter**. FGT-EDGE blocks the website.
2. Return to **FGT-EDGE**.
3. Click **Log & Report > Web Filter**.
4. The web filter log contains an entry for when FGT-EDGE applied web filtering to block Twitter.

| Date/Time           | User | Source       | Action  | URL                             | Category Description | Initiator | Sent/Re |
|---------------------|------|--------------|---------|---------------------------------|----------------------|-----------|---------|
| 2021/01/04 13:38:17 |      | 172.16.20.51 | blocked | https://twitter.com/favicon.ico | Social Networking    |           | 337 B/O |
| 2021/01/04 13:38:17 |      | 172.16.20.51 | blocked | https://twitter.com/            | Social Networking    |           | 409 B/O |

5. Select the log entry and click **Details** to view more information.



## Stop and Think

This exercise used local logging, with logs stored directly on FGT-EDGE. What reasons could AcmeCorp have for using external logging, such as FortiGate Cloud or a FortiAnalyzer, instead of local logging?

### ----- Hint 1 Section -----

**Hint: 1**                      **Points: 2**

**Hint Text:**

## Hint

FortiGate Cloud is a cloud-based infrastructure management and log retention service offered by Fortinet. You can use it for both analytics and management of multiple FortiGate devices, as well as FortiSwitch and FortiAP devices.

FortiAnalyzer offers advanced logging and reporting capabilities, centralized security analytics across the Fortinet Security Fabric, and security automation via Fabric Connectors and

application programming interfaces (APIs). These use cases enable security teams to increase efficiency, reduce risk, and improve total cost of ownership (TCO).

----- **Answer Section** -----

**Answer:** radio

**Answer Text:**

**Answer**

Correct: D

All three reasons would require the use of external logging.

**Answer Key:**

- ☐ 1. AcmeCorp is using a FortiGate model that doesn't have a hard drive/storage space for logs
- ☐ 2. AcmeCorp needs to aggregate logs from multiple FortiGate devices
- ☐ 3. AcmeCorp requires remote access to logs
- ☒ 4. All of the above

**Index:** 5.0 (d)

**Use Case:** Logging and Reporting

**Objective Title:** Learn More About Logging and Reporting

**Points:** 0

----- **Objective Section** -----

**Objective Text:**

## Learn More About Logging and Reporting

If you would like to learn more about logging and reporting, as your instructor about the *Simplify SOC Operations for the Security Fabric with FortiAnalyzer* course offered by the Fast Track Program.

FortiAnalyzer, part of the Fortinet Security Fabric, address the complexity of operations that security teams around the world face. FortiAnalyzer enables an organization to maximize the impact and effectiveness of a lean security team. It does this by providing broad visibility and control of an organization's entire digital attack surface, an integrated solution reducing the complexity of supporting multiple point products, and automating of security workflows increases the speed of operation.

Participants who attend this workshop will learn how to:

- Understand the benefits of using FortiAnalyzer to simplify SOC operations.
- How to use playbooks to automate workflows in order reduce the workload on the security team.
- How to use FortiGate event handlers to automate actions via automation stitches.
- How to work with analytics logs and generate custom reports

**Index:** 6.0

**Use Case:** CLI Basics

**Objective Title:** New Objective

**Points:** 0

----- **Objective Section** -----

**Objective Text:**

## CLI Basics

In this lab, you use the FortiGate CLI to add security profiles and logging to the DC firewall policy. You also learn about tips for using the CLI.

The steps you complete to accomplish this lab are:

1. Connect to the CLI
2. Use basic commands
3. Configure the DC firewall policy
4. Examine traffic with packet sniffing

Time to complete: 15 minutes

**Index:** 6.0 (a)

**Use Case:** CLI Basics

**Objective Title:** Connect to the CLI

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

## Connect to the CLI

### Background

In this exercise, you connect to the FortiGate three ways:

- The CLI console button in the GUI
- The drop-down menu available in the GUI for certain objects
- Connecting via SSH using a terminal emulator application such as PuTTY

### Tasks

#### CLI Console Button

1. Connect to FGT-EDGE.
2. Click >\_ in the top right-hand corner to connect to the CLI console.



3. Type the following command `get system status` and press Enter.
4. An output similar to the screenshot below appears, listing information about the FortiGate:

```
CLI Console (1)

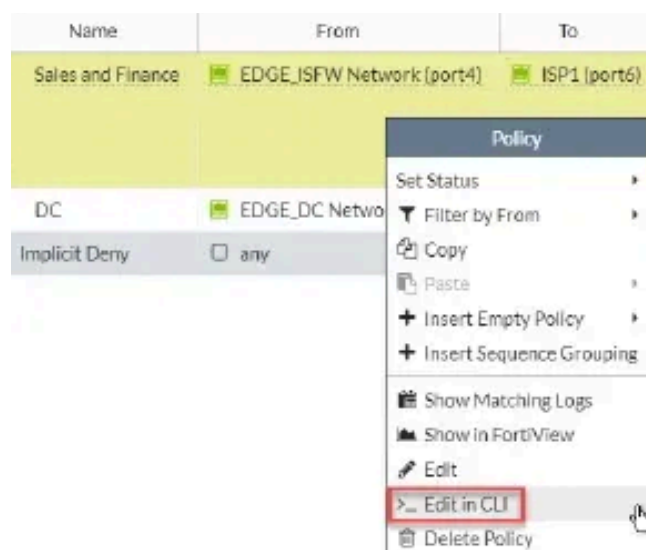
FGT-EDGE # get system status
Version: FortiGate-VM64 v6.4.2,build1723,200730 (GA)
Virus-DB: 79.00882(2020-08-25 04:20)
Extended DB: 79.00882(2020-08-25 04:20)
Extreme DB: 1.00000(2018-04-09 18:07)
IPS-DB: 6.00741(2015-12-01 02:30)
IPS-ETDB: 15.00910(2020-08-21 00:34)
APP-DB: 15.00909(2020-08-20 00:05)
INDUSTRIAL-DB: 15.00910(2020-08-21 00:34)
Serial-Number: FGV01TM19002139
IPS Malicious URL Database: 2.00746(2020-08-24 06:48)
License Status: Valid
VM Resources: 1 CPU/1 allowed, 2010 MB RAM
Log hard disk: Available
Hostname: FGT-EDGE
Operation Mode: NAT
Current virtual domain: root
Max number of virtual domains: 10
Virtual domains status: 1 in NAT mode, 0 in TP mode
Virtual domain configuration: disable
FIPS-CC mode: disable
Current HA mode: standalone
Branch point: 1723
Release Version Information: GA
FortiOS x86-64: Yes
System time: Wed Oct 21 13:32:04 2020

FGT-EDGE #
```

5. Close the CLI console screen.

## Drop-Down Menu

1. Click **Policy & Objects > Firewall Policy**, right-click **Sales and Finance**, and use the drop-down menu to select **Edit in CLI**.



2. The CLI console opens and shows the configuration for the firewall policy:

```

FGT-EDGE # config firewall policy

FGT-EDGE (policy) # edit 5

FGT-EDGE (5) # show
config firewall policy
    edit 5
        set name "Sales and Finance"
        set uuid bc2eb760-d1c1-51ea-f753-12b3d98a183e
        set srcintf "port4"
        set dstintf "port6"
        set srcaddr "Sales and Finance"
        set dstaddr "all"
        set action accept
        set schedule "always"
        set service "ALL"
        set utm-status enable
        set ssl-ssh-profile "deep-inspection"
        set av-profile "Sales and Finance"
        set webfilter-profile "Sales and Finance"
        set application-list "Sales and Finance"
        set groups "Finance" "Sales" "Guests"
        set nat enable
    next
end

```

3. Minimize Chrome.

## Using SSH

1. Return to the Lab Activity Tab. Click **FGT-EDGE** in the sidebar menu under the Core group, and then click on the **SSH** option

**Note:** This view is the same view you would have had you connected to FGT-EDGE using a terminal emulator application such as PuTTY.

2. You are connected to the CLI.

**Note:** You did not have to provide credentials because the NSE Institute Portal automatically passes the admin credentials to the FortiGate.

3. Type the following command: `get system interface physical` and press Enter.
4. An output similar to the screenshot below appears, listing information about the FortiGate physical interfaces:

```
FGI-EDGE # get system interface physical
== [onboard]
  ==[port1]
    mode: static
    ip: 192.168.0.101 255.255.255.0
    ipv6: ::/0
    status: up
    speed: 1000Mbps (Duplex: full)
  ==[port2]
    mode: static
    ip: 10.10.30.30 255.255.255.240
    ipv6: ::/0
    status: up
    speed: 1000Mbps (Duplex: full)
  ==[port3]
    mode: static
    ip: 10.10.30.6 255.255.255.248
    ipv6: ::/0
    status: up
    speed: 1000Mbps (Duplex: full)
  ==[port4]
    mode: static
    ip: 10.10.30.14 255.255.255.248
--More--
```

**Index:** 6.0 (b)

**Use Case:** CLI Basics

**Objective Title:** Use Basic Commands

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

## Use Basic Commands

### Background

In the previous objectives, you use the following CLI commands:

- `get`: Get information about the current configuration
- `config`: Configure objects and system settings
- `execute`: Execute static commands

### Tasks

#### Using the `get` Command

1. Return to your browser tab with the GUI interface to the **FGT-EDGE**.
2. Click `>_` to connect to the CLI console.
3. To view the `get` commands, type `get ?`
4. The commands shown are the top level commands. Additional sub-commands are available for some of these commands, such as `system`.

```

FGT-EDGE # get
alertemail      Alert email configuration.
antivirus        AntiVirus configuration.
application      Application control configuration.
authentication    authentication
cifs             cifs
credential-store credential-store
dlp              DLP configuration.
dnfilter         DNS filter configuration.
dpdk             FortiOS DPDK Helper configuration.
emailfilter      AntiSpam configuration.
endpoint-control Endpoint control configuration.
extender         extender
extender-controller FortiExtender Controller configuration.
file-filter      file-filter
firewall         Firewall configuration.
ftp-proxy        FTP proxy configuration.
hardware         hardware
icap             ICAP client configuration.
ips              IPS configuration.
ipsec            ipsec
log              Log configuration.
mgmt-data        mgmt-data
report           Report configuration.
router           Router configuration.
ssh-filter       SSH filter configuration.
switch-controller External FortiSwitch configuration.
system           System operation configuration.
test             test
user             Authentication configuration.
voip             VoIP configuration.
vpn              VPN configuration.
waf              Web Application Firewall configuration.
wanopt           WAN optimization configuration.
web-proxy        Web proxy configuration.
webfilter        Web filter configuration.
wireless-controller Wireless access point configuration.

```

5. To view the get system sub-commands, type get system ?

```

FGT-EDGE # get system
acccprofile      Configure access profiles for
system administrators.
admin            Configure admin users.
affinity-interrupt Configure interrupt affinity.
affinity-packet-redistribution.
alias            Configure alias command.
api-user         Configure API users.
arp              IPv4 ARP table.
arp-table        Configure ARP table.
auto-install     Configure USB auto installation.
auto-script      Configure auto script.
auto-update      auto-update
automation-action Action for automation stitches.
automation-destination Automation destinations.
automation-stitch Automation stitches.
automation-trigger Trigger for automation stitches.
autoupdate       Configure automatic updates.
central-management Configure central management.
central-mgmt     Configuration of Central Management Service.
checksum         checksum
cluster-sync     Configure FortiGate Session Life Support Protocol (FGSLP) session synchronization.
cmdb             System CMDB information.
console         Configure console.
csf              Add this FortiGate to a Security Fabric or set up a new Security Fabric on this FortiGate.
custom-language  Configure custom languages.
ddns             Configure DDNS.
dhcp             Configure DHCP.
dhcp6            Configure DHCPv6.
dns              Configure DNS.
dns-database     Configure DNS databases.
dns-server       Configure DNS servers.
dscp-based-priority table.
email-server     Configure the email server used by the FortiGate various things. For example, for sending email messages to users to support user authentication features.
external-resource Configure external resource.
fips-cc          Configure FIPS-CC mode.

```

6. To view the list of administrative users, type `get system admin` and press Enter. The output lists all administrative users:

```
FGT-EDGE # get system admin
== [ admin ]
name: admin
== [ bob-admin ]
name: bob-admin
```

## Using the `config` Command

1. In this exercise, you edit the bob-admin account. This account can't be edited if the user is logged in. To check if bob-admin is logged in, click **Dashboard** > **Status** and locate the **Administrators** widget. If **bob-admin** appears on the list, click the widget and select **Show active administrator sessions**.



2. Select **bob-admin** and click **Disconnect**.
3. Click **>\_** to connect to the CLI console.
4. To view the `config` commands, type `config ?`

```
FGT-EDGE # config
alertemail      Alert email configuration.
antivirus        AntiVirus configuration.
application      Application control configuration.
authentication    authentication
cifs             cifs
credential-store credential-store
dlp              DLP configuration.
dnsmfilter       DNS filter configuration.
dpdk             FortiOS DPDK Helper configuration.
emailfilter      AntiSpam configuration.
endpoint-control Endpoint control configuration.
extender-controller FortiExtender Controller configuration.
file-filter      file-filter
firewall         Firewall configuration.
ftp-proxy        FTP proxy configuration.
icap             ICAP client configuration.
ips             IPS configuration.
log             Log configuration.
report           Report configuration.
router           Router configuration.
ssh-filter       SSH filter configuration.
switch-controller External FortiSwitch configuration.
system           System operation configuration.
user            Authentication configuration.
voip            VoIP configuration.
vpn             VPN configuration.
waf             Web Application Firewall configuration.
wanopt          WAN optimization configuration.
web-proxy        Web proxy configuration.
webfilter        Web filter configuration.
wireless-controller Wireless access point configuration.
```

5. To configure administrative accounts, type `config system admin` and press Enter.
6. Type `edit bob-admin` to configure Bob's administrator account and press Enter.
7. Type `show` and press Enter to view the current configuration of the account.

```
FGT-EDGE # config system admin
FGT-EDGE (admin) # edit bob-admin
FGT-EDGE (bob-admin) # show
config system admin
  edit "bob-admin"
    set trusthost1 172.16.20.51 255.255.255.255
    set accprofile "super_admin_readonly"
    set vdom "root"
    set password ENC SH2s/Hymt6L7t8wjGMzyKh+1UaT+xz1fUK/K1GRabJe/Dsp9o0147u1rrZWkzI=
  next
end
```

8. Type `set accprofile super_admin` and press Enter to change the administrative profile.
9. Type `end` and press Enter to save the configuration changes.

```
FGT-EDGE (bob-admin) # set accprofile super_admin
FGT-EDGE (bob-admin) # end
```

10. Close the CLI console.
11. To confirm the change, click **System > Administrators**. The **Profile** for **bob-admin** is now **super\_admin**.

| Name                   | Trusted Hosts   | Profile     |
|------------------------|-----------------|-------------|
| System Administrator 3 |                 |             |
| admin                  |                 | super_admin |
| bob-admin              | 172.16.20.51/32 | super_admin |

## Using the execute Command

1. Click `>_` to connect to the CLI console.
2. To view the execute commands, type `execute ?`

```

FGT-EDGE # execute
api-user          api-user
auto-script       auto-script
backup            backup
batch             batch
central-mgmt      central-mgmt
cfg               cfg
clear             clear
cli               cli
config-transaction config-transaction
cpu               cpu
date              System date.
dhcp              dhcp
dhcp6             dhcp6
disconnect-admin-session Disconnect logged-in admin user.
disk              disk
enter             Select virtual domain.
erase-disk         Erase a disk. [Take 0-1 arg(s)]
extender           extender
factoryreset      Reset to factory default now.
factoryreset-shutdown Reset to factory default and shutdown.
factoryreset2      Reset to factory default except system.global.vdom-mode/VDOMs/system.interface/system.settings/router.static/router.static6.
fctema            fctema
firewall           firewall
formatlogdisk      Format log disk.
forticarrier-license Forticarrier-license
forticloud-sandbox forticloud-sandbox
fortiguard-log     fortiguard-log
fortiguard-message fortiguard-message
fortitoken         fortitoken
fortitoken-cloud   fortitoken-cloud
fortitoken-mobile   fortitoken-mobile
fsso               Fortinet Single Sign-On.
ha                 High Availability.
interface          interface
internet-service   internet-service
log                log
mrouter            mrouter
nsx                 NSX management.
ping               PING command.
ping-options       ping-options
--More--

```

3. To run a traceroute to the Fortinet website, type `execute traceroute www.fortinet.com` and press Enter.
4. After a few moments, an output appears similar to the following screenshot:

```
FGT-EDGE # execute traceroute www.fortinet.com
traceroute to www.fortinet.com (13.56.33.144), 32 hops max, 3 probe packets per hop, 84 byte packets
 1 100.65.0.254 0.543 ms 0.703 ms 0.384 ms
 2 100.64.1.1 0.469 ms 0.434 ms 0.308 ms
 3 172.16.81.254 0.461 ms 0.459 ms 0.359 ms
 4 172.16.15.254 0.429 ms 0.536 ms 0.376 ms
 5 206.47.132.1 0.482 ms 0.621 ms 0.439 ms
 6 96.45.47.75 0.555 ms 0.985 ms 0.681 ms
 7 184.94.115.129 3.878 ms 3.976 ms 3.620 ms
 8 64.125.14.234 <ae1.mpr1.yul1.ca.zip.zayo.com> 3.495 ms 3.746 ms 3.447 ms
 9 64.125.27.163 <ae8.mpr2.yul2.ca.zip.zayo.com> 3.749 ms 3.733 ms 3.585 ms
10 * * 64.125.27.158 <ae9.cs4.ord2.us.zip.zayo.com> 64.109 ms
11 64.125.29.28 <ae6.cs2.sea1.us.eth.zayo.com> 62.868 ms 61.714 ms 83.894 ms
12 64.125.29.3 <ae27.mpr2.sea1.us.zip.zayo.com> 61.456 ms 61.398 ms 62.798 ms
13 99.82.182.182 61.689 ms 61.589 ms 61.578 ms
14 * * *
15 * * *
16 * * *
17 54.239.42.238 75.144 ms 74.715 ms 75.086 ms
18 52.93.131.223 76.683 ms 76.399 ms 75.973 ms
19 * * *
20 54.248.242.114 77.689 ms 76.988 ms 77.128 ms
21 54.248.242.189 74.154 ms 79.215 ms 74.167 ms
22 * * *
23 * * *
24 72.21.222.17 83.843 ms 83.754 ms 85.121 ms
25 * * *
26 * * *
27 * * *
28 * * *
29 * * *
30 * * *
31 13.56.33.144 <www.fortinet.com> 79.862 ms 79.566 ms 79.548 ms
```

**Index:** 6.0 (c)

**Use Case:** CLI Basics

**Objective Title:** Configure the DC Firewall Policy

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

## Configure the DC Firewall Policy

### Background

In the previous objectives, you used the GUI to configure the Sales and Finance policy to include security profiles and logging. In this exercise, you use the CLI to configure the DC policy to include security profiles and logging. The security profiles you apply have been pre-configured for use with the DC network.

### Tasks

1. Click >\_ to connect to CLI console.
2. Enter the following commands to add security profiles and logging:

```
config firewall policy
  edit 2
    set utm-status enable
    set ssl-ssh-profile deep-inspection
    set av-profile DC
    set webfilter-profile DC
    set application-list DC
    set logtraffic all
  end
```

```

FGT-EDGE # config firewall policy

FGT-EDGE (policy) # edit
policyid      Policy ID (0 - 4294967294).
5
2

FGT-EDGE (policy) # edit 2
FGT-EDGE (2) # set utm-status enable

FGT-EDGE (2) # set ssl-ssh-profile deep-inspection

FGT-EDGE (2) # set av-profile DC

FGT-EDGE (2) # set webfilter-profile DC

FGT-EDGE (2) # set application-list DC

FGT-EDGE (2) # set logtraffic all

FGT-EDGE (2) # end

```

3. Close the CLI console.
4. Click **Policy & Objects > Firewall Policy**. You can see the changes made to the **DC** policy under **Security Profiles** and **Log**.

| Name              | From                    | To           | Source                                           | Destination | Schedule | Service | Action | NAT     | Security Profiles                                                              | Log |
|-------------------|-------------------------|--------------|--------------------------------------------------|-------------|----------|---------|--------|---------|--------------------------------------------------------------------------------|-----|
| Sales and Finance | EDGE_SFWNetwork [port0] | ISP1 [port0] | Finance<br>Sales<br>Clients<br>Sales and Finance | all         | always   | ALL     | ACCEPT | Enabled | Sales and Finance<br>Sales and Finance<br>Sales and Finance<br>deep-inspection | All |
| DC                | EDGE_DCNetwork [port0]  | ISP1 [port0] | DC                                               | all         | always   | ALL     | ACCEPT | Enabled | DC<br>DC<br>DC<br>deep-inspection                                              | All |

**Index:** 6.0 (d)

**Use Case:** CLI Basics

**Objective Title:** Examine Traffic With Packet Sniffing

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

## Examine Traffic With Packet Sniffing

### Background

Packet sniffing allows you to gather information about specific packets passing through the AcmeCorp network to help troubleshoot issues. To do this, you use the `diagnose sniffer` CLI command.

The full syntax for the packet sniffing command is `diagnose sniffer packet <interface> <filter> <verbose> <count> <tsformat>`. The parts of the command are as follows:

- `<interface>`: an interface name or any for all interfaces
- `<filter>`: the selected filter
- `<verbose>`: the level of verbosity
- `<count>`: the number of packets the sniffer reads before stopping
- `<tsformat>`: the timestamp format

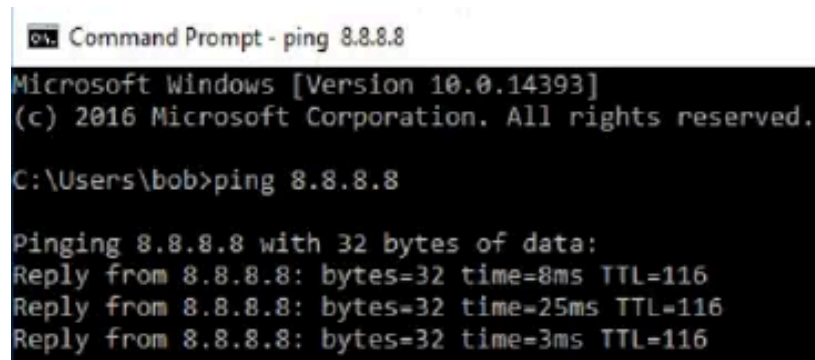
Only the interface value is required.

In the exercise, you use packet sniffing as part of the troubleshooting process and examine traffic on any interface. You use the filter `'host 8.8.8.8 and icmp'` to look for ICMP traffic from the host IP address 8.8.8.8 (a Google public DNS server). You use verbose level 4, to print the header of the packets with the interface name listed. You set count to 100 and use the 1 timestamp format, to view absolute local time in yyyy-mm-dd hh:mm:ss.ms format.

### Tasks

1. Click `>_` to connect to CLI console.
2. Enter the command `diagnose sniffer packet any 'host 8.8.8.8 and icmp' 4 100 1` and press Enter.

3. Return to **Bob's Computer**.
4. Open Command Prompt.
5. Type `ping 8.8.8.8` and press Enter.

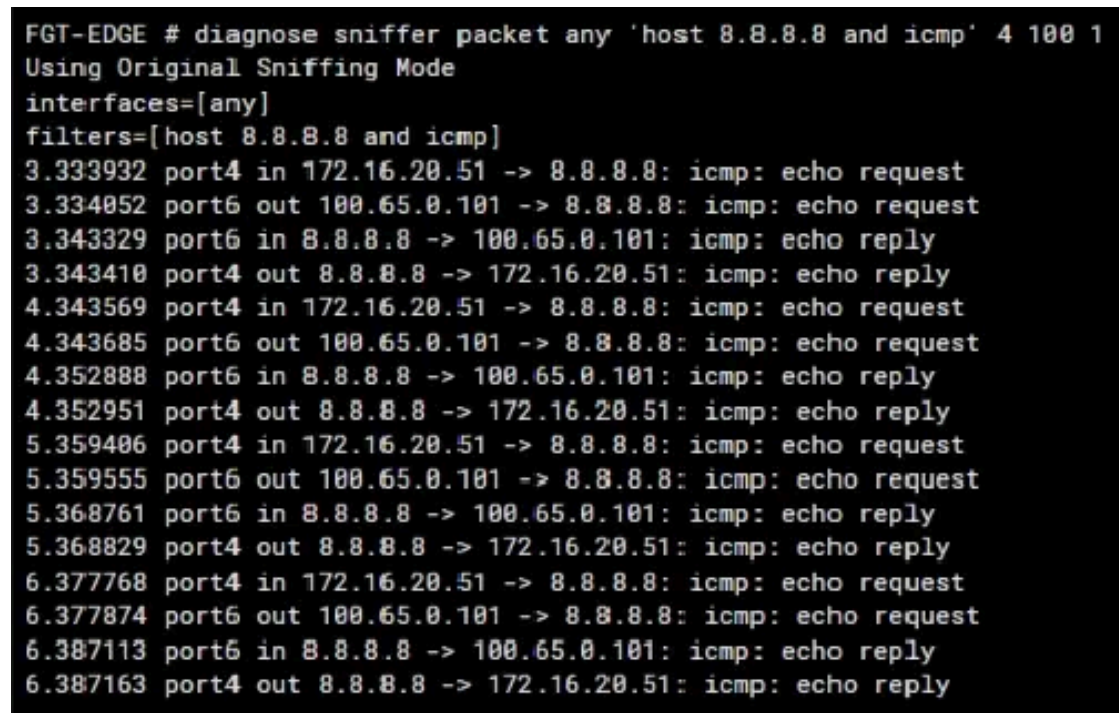


```
Command Prompt - ping 8.8.8.8
Microsoft Windows [Version 10.0.14393]
(c) 2016 Microsoft Corporation. All rights reserved.

C:\Users\bob>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=8ms TTL=116
Reply from 8.8.8.8: bytes=32 time=25ms TTL=116
Reply from 8.8.8.8: bytes=32 time=3ms TTL=116
```

6. Return to the **FGT-EDGE** and view the CLI console.
7. An output similar to the following appears, showing information about the packets generated by Bob's computer. In order to exit out of this, press `Ctrl + C`.



```
FGT-EDGE # diagnose sniffer packet any 'host 8.8.8.8 and icmp' 4 100 1
Using Original Sniffing Mode
interfaces=[any]
filters=[host 8.8.8.8 and icmp]
3.333932 port4 in 172.16.20.51 -> 8.8.8.8: icmp: echo request
3.334052 port6 out 100.65.0.101 -> 8.8.8.8: icmp: echo request
3.343329 port6 in 8.8.8.8 -> 100.65.0.101: icmp: echo reply
3.343410 port4 out 8.8.8.8 -> 172.16.20.51: icmp: echo reply
4.343569 port4 in 172.16.20.51 -> 8.8.8.8: icmp: echo request
4.343685 port6 out 100.65.0.101 -> 8.8.8.8: icmp: echo request
4.352888 port6 in 8.8.8.8 -> 100.65.0.101: icmp: echo reply
4.352951 port4 out 8.8.8.8 -> 172.16.20.51: icmp: echo reply
5.359406 port4 in 172.16.20.51 -> 8.8.8.8: icmp: echo request
5.359555 port6 out 100.65.0.101 -> 8.8.8.8: icmp: echo request
5.368761 port6 in 8.8.8.8 -> 100.65.0.101: icmp: echo reply
5.368829 port4 out 8.8.8.8 -> 172.16.20.51: icmp: echo reply
6.377768 port4 in 172.16.20.51 -> 8.8.8.8: icmp: echo request
6.377874 port6 out 100.65.0.101 -> 8.8.8.8: icmp: echo request
6.387113 port6 in 8.8.8.8 -> 100.65.0.101: icmp: echo reply
6.387163 port4 out 8.8.8.8 -> 172.16.20.51: icmp: echo reply
```

8. The output shows both the ingress and egress interfaces on FGT-EDGE that ICMP traffic flows through to reach 8.8.8.8 from Bob's computer.

**Index:** 7.0

**Use Case:** Local User Authentication

**Objective Title:** New Objective

**Points:** 0

----- **Objective Section** -----

**Objective Text:**

## Local User Authentication

In this lab, you configure local user and device authentication to provide greater visibility into the users and devices on the AcmeCorp network. You also configure FGT-EDGE to create and manage temporary guest accounts.

The steps you complete to accomplish this lab are:

1. Create user groups and accounts
2. Add authentication to the firewall policy
3. Manage administrator accounts
4. Configure guest accounts

Time to complete: 20 minutes

**Index:** 7.0 (a)  
**Use Case:** Local User Authentication  
**Objective Title:** Create User Groups and Accounts  
**Points:** 10

----- **Objective Section** -----  
**Objective Text:**

## Create User Groups and Accounts

### Background

In this exercise, you will be working on the FGT-EDGE to create two user groups: Sales and Finance. Then you create user accounts for Alice and Bob, adding them to the appropriate group.

### Tasks

1. Click **User & Authentication > User Groups** and click **Create New**.
2. Set **Name** to Sales.
3. Set **Type** to **Firewall**.

|         |                                                                                                                                |
|---------|--------------------------------------------------------------------------------------------------------------------------------|
| Name    | <input type="text" value="Sales"/>                                                                                             |
| Type    | <div><div>Firewall</div><div>Fortinet Single Sign-On (FSSO)</div><div>RADIUS Single Sign-On (RSSO)</div><div>Guest</div></div> |
| Members | <div><div></div><div>+</div></div>                                                                                             |

4. Click **OK**.
5. Repeat the above steps to create a second group named Finance.

|         |                                                                                                                                |
|---------|--------------------------------------------------------------------------------------------------------------------------------|
| Name    | <input type="text" value="Finance"/>                                                                                           |
| Type    | <div><div>Firewall</div><div>Fortinet Single Sign-On (FSSO)</div><div>RADIUS Single Sign-On (RSSO)</div><div>Guest</div></div> |
| Members | <div><div></div><div>+</div></div>                                                                                             |

6. Click **User & Authentication > User Definition** and click **Create New**.

7. Set **User Type** to **Local User**.



The screenshot shows a horizontal progress bar with four steps: 1 User Type, 2 Login Credentials, 3 Contact Info, and 4 Extra Info. Below the progress bar is a dropdown menu for 'Local User'. The dropdown list includes the following options: Local User, Remote RADIUS User, Remote TACACS+ User, Remote LDAP User, FSSO, FortiClient EMS User, and FortiNAC User.

8. Click **Next**.

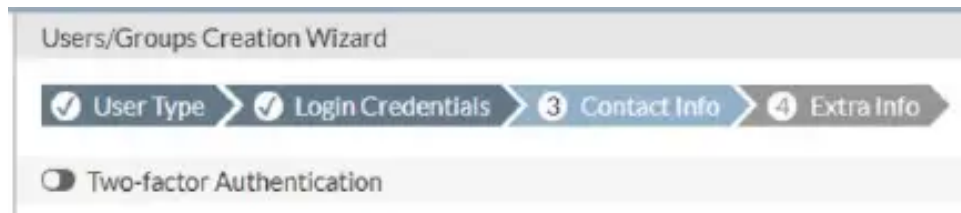
9. Set **Username** to **alice** and **Password** to **Fortinet1!**



The screenshot shows the 'Login Credentials' step of the wizard. The progress bar indicates steps 1 User Type, 2 Login Credentials, 3 Contact Info, and 4 Extra Info. Below the progress bar, the 'Username' field contains the text 'alice' and the 'Password' field contains a series of dots representing a masked password.

10. Click **Next**.

11. Leave Two-factor Authentication disabled.



The screenshot shows the 'Contact Info' step of the wizard. The progress bar indicates steps 1 User Type, 2 Login Credentials, 3 Contact Info, and 4 Extra Info. Below the progress bar, there is a section for 'Two-factor Authentication' with a radio button that is currently unselected.

12. Click **Next**.

13. Turn on **User Group** and select **Sales**.



The screenshot shows the 'Extra Info' step of the wizard. The progress bar indicates steps 1 User Type, 2 Login Credentials, 3 Contact Info, and 4 Extra Info. Below the progress bar, there are two fields: 'User Account Status' with a dropdown menu showing 'Enabled' and 'Disabled', and 'User Group' with a dropdown menu showing 'Sales'.

14. Click **Submit**.

15. Click **Create New** to create a second user account.

16. Set **User Type** to **Local User**.

1 User Type > 2 Login Credentials > 3 Contact Info > 4 Extra Info

|                      |
|----------------------|
| Local User           |
| Remote RADIUS User   |
| Remote TACACS+ User  |
| Remote LDAP User     |
| FSSO                 |
| FortiClient EMS User |
| FortiNAC User        |

17. Click **Next**.

18. Set **Username** to bob and **Password** to Fortinet1!

1 User Type > 2 Login Credentials > 3 Contact Info > 4 Extra Info

Username

Password

19. Click **Next**.

20. Leave Two-factor Authentication disabled.

Users/Groups Creation Wizard

1 User Type > 2 Login Credentials > 3 Contact Info > 4 Extra Info

☐ Two-factor Authentication

21. Click **Next**.

22. Turn on **User Group** and select **Finance**.

1 User Type > 2 Login Credentials > 3 Contact Info > 4 Extra Info

User Account Status ☒ Enabled ☐ Disabled

User Group ☒

23. Click **Submit**.

**Index:** 7.0 (b)

**Use Case:** Local User Authentication

**Objective Title:** Add Authentication to the Firewall Policy

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

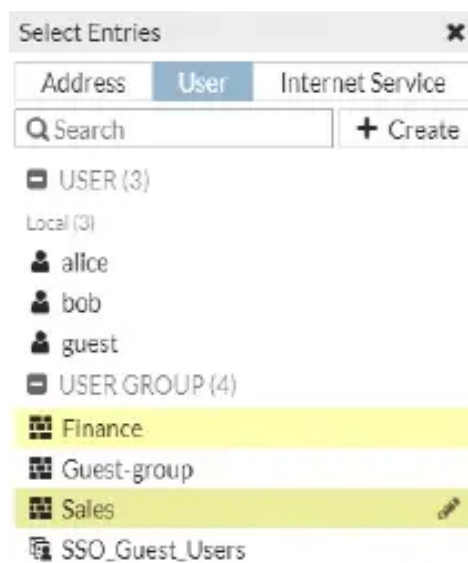
## Add Authentication to the Firewall Policy

### Background

In this exercise, you add user authentication to the firewall policy **Sales and Finance**.

### Tasks

1. Click **Policy & Object > Firewall Policy**, click **Sales and Finance**, and click **Edit**.
2. Click **Source**. In the right-hand menu, click **User** and select both **Finance** and **Sales**.



3. Click **Close**.
4. Click **OK**.
5. Return to **Bob's Computer**.
6. Open Chrome and select the browser bookmark **Google**.
7. An authentication screen appears in a new tab. Enter the credentials for the bob account and click **Continue**.



## Authentication Required

Please enter your username and password to continue.

Username

Password

Continue

- Once your authentication is complete, you can access the website.
- Return to **FGT-EDGE**.
- Click **Dashboard > Users & Devices** and click the **Firewall Users** widget to expand it. **bob** appears on the list of connected users.



**Index:** 7.0 (c)  
**Use Case:** Local User Authentication  
**Objective Title:** Manage Administrator Accounts  
**Points:** 10

----- **Objective Section** -----  
**Objective Text:**

## Manage Administrator Accounts

### Background

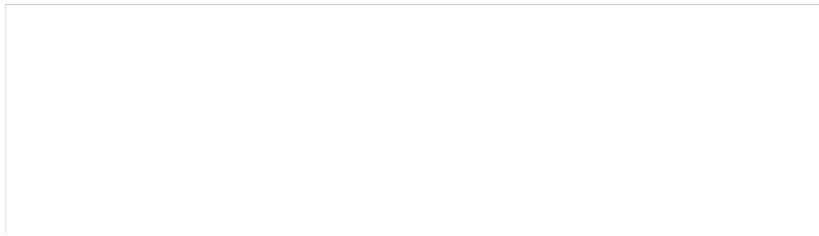
In this exercise, you create a new administrator account for Bob, an AcmeCorp employee. This account configuration includes a trusted host, so Bob can only access it from her assigned computer on the network.

### Tasks

1. Click **System > Administrators** and use the **Create New** dropdown menu to select **Administrator**.
2. Configure the following settings:
  - **Username:** bob-admin
  - **Type:** Local User
  - **Password:** Fortinet1!
  - **Confirm Password:** Fortinet1!
  - **Administrator Profile:** super\_admin\_readonly

|                       |                                                                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Username              | <input type="text" value="bob-admin"/>                                                                                                                                                  |
| Type                  | <div><div>Local User</div><div>Match a user on a remote server group</div><div>Match all users in a remote server group</div><div>Use public key infrastructure (PKI) group</div></div> |
| Password              | <input type="password" value="••••••••"/>                                                                                                                                               |
| Confirm Password      | <input type="password" value="••••••••"/>                                                                                                                                               |
| Comments              | <div><input type="text" value="Write a comment..."/></div> <div>0/255</div>                                                                                                             |
| Administrator Profile | <div>super_admin_readonly</div>                                                                                                                                                         |
| Email Address         | <input type="text"/>                                                                                                                                                                    |

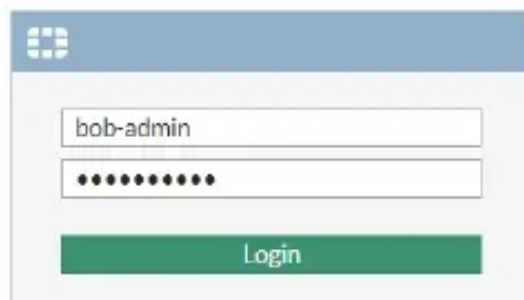
3. Turn on **Restrict login to trusted hosts**.
4. Set **Trusted Host 1** to 172.16.20.51/32, the IP address of Bob’s computer.



5. Click **OK**.
6. Use the dropdown menu in the top right of the screen to select **Logout**.



7. Enter the credentials for the bob-admin account.



8. The authentication attempt fails because the trusted host doesn't match.
9. Return to **Bob's Computer**.
10. Run Chrome and click the browser bookmark **FGT-EDGE**.
11. Enter the credentials for the **bob-admin** account.
12. When the message about FortiGate Setup appears, click **Later**.



13. You can now access the FortiGate GUI with this account.

## Stop and Think

While logged in using the bob-admin account, click **System** > **Settings**. You can view the system settings but aren't able to change any of them. Why is this?

### ----- Hint 1 Section -----

**Hint: 1**                      **Points: 2**

**Hint Text:**

## Hint

Administrator profiles define what the administrator can do when logged into the FortiGate. There are several default profiles you can use, plus you can create new ones as required.

The default profiles are:

- **super\_admin**: allows full read-write access to the entire FortiGate configuration
- **prof\_admin**: allows read-write access to most of the FortiGate configuration, except for routing, system settings, and endpoint control
- **super\_admin\_readonly**: allows read-only access to the entire FortiGate configuration
- **prof\_admin\_readonly**: allows read-only access to most of the FortiGate configuration, except for routing, system settings, and endpoint control

To create, edit, and delete profiles, click **System** > **Admin Profiles**. You can't delete the **super\_admin** profile.

**Note:** The read-only versions of the default admin profiles don't appear on this page.

### ----- Answer Section -----

**Answer:** radio

**Answer Text:**

**Answer**

Correct: C

The bob account uses the **super\_admin\_readonly** admin profile, so when you are logged in with this account, you can't change any settings.

The default admin account used the **super\_admin** admin profile, allowing you full read-write access to the configuration when you log in with this account.

**Answer Key:**

- ☒ 1. The bob-admin account was configured incorrectly
- ☒ 2. Only the default admin account can change system settings
- ☒ 3. The administrator profile for the bob-admin account is restricted to read-only access
- ☒ 4. You can't access system settings when a trusted host is required

**Index:** 7.0 (d)  
**Use Case:** Local User Authentication  
**Objective Title:** Configure Guest Accounts  
**Points:** 10

----- **Objective Section** -----  
**Objective Text:**

## Configure Guest Accounts

### Background

In this exercise, you create a guest user group, then configure the settings to create temporary guest accounts.

### Tasks

1. Click **User & Authentication > User Groups** and click **Create New**.
2. Set **Name** to `Guests`.
3. Set **Type** to **Guest**.

Name:

Type: 

Firewall  
Fortinet Single Sign-On (FSSO)  
RADIUS Single Sign-On (RSSO)  
**Guest**

Batch Guest Account Creation: ☐

User ID: 

Email | Auto Generated | Specify

Maximum Accounts: ☐

4. Under **Guest Details**, set **Password** to **Specify**.

Guest Details

Enable Name: ☐

Enable Email: ☒

Enable SMS: ☐

Password: ☒

Auto Generated | **Specify**

Sponsor: ☒

Optional | Required

Company: ☒

Optional | Required

5. Under **Expiration**, set **Start Countdown** to **After First Login** and **Time** to 30 seconds.

**Note:** The password and expiration settings you use in this exercise are for testing purposes.

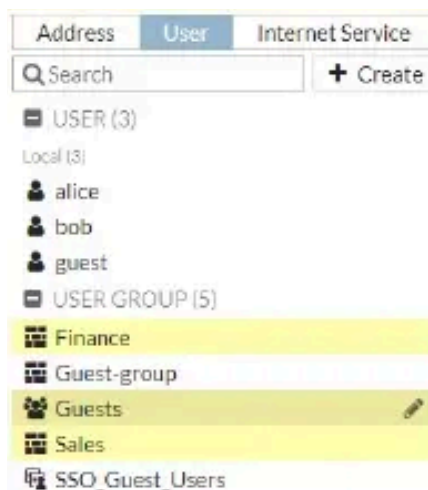
- 6. Leave all other settings at the default.
- 7. Click **OK**.
- 8. Click **User & Authentication > Guest Management** and click **Create New**.
- 9. Set **Password** to Fortinet1! and **Email** to guest@acmecorp.net.

|            |                    |
|------------|--------------------|
| User ID    | Use Email Address  |
| Password   | Fortinet1!         |
| Sponsor    |                    |
| Company    |                    |
| Email      | guest@acmecorp.net |
| Expiration | 30 Seconds         |
| Comments   |                    |

- 10. Click **OK**.
- 11. Click **Policy & Objects > Firewall Policy**. For the **Sales and Finance** policy, hover over the **Source** column and then click the edit icon.

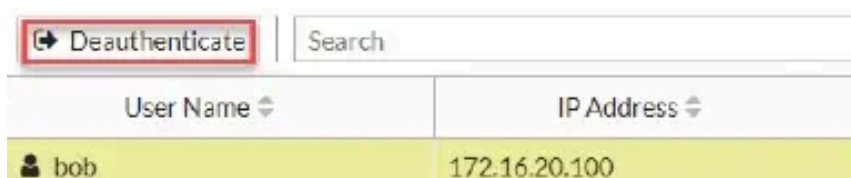
| Source            |
|-------------------|
| Finance           |
| Sales             |
| Sales and Finance |

- 12. In the right-hand list, click **User** and select **Guests**.



13. Click **Apply**.

14. Click **Dashboard > Users & Devices** and expand the **Firewall Users** widget. Select **bob** and click **Deauthenticate**.

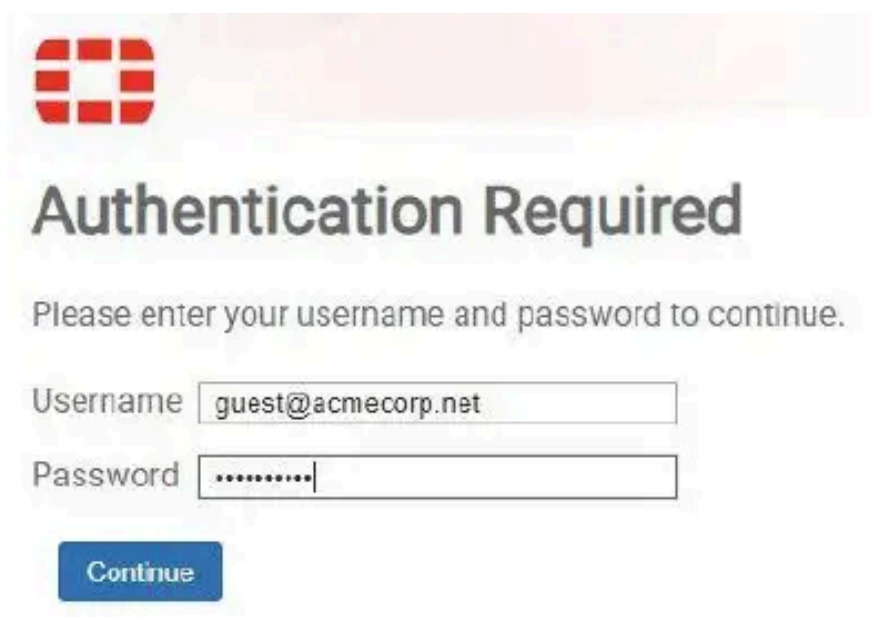


15. Click **OK**.

16. Return to **Bob's Computer**.

17. Open Chrome and select the browser bookmark **Fortinet**.

18. An authentication screen appears. Enter the credentials for the guest account.



19. Once your authentication is complete, you can access the website.

- 20. Return to the **FGT-EDGE**.
- 21. Click **Dashboard > Users & Devices**, and expand the **Firewall Users** widget.
- 22. If **guest@acmecorp.net** appears on the list, select it, click **Deauthenticate**, then click **OK**.
- 23. Return to **Bob’s Computer** and click **Fortinet** again. When prompted, enter the credentials for the guest account.
- 24. Your authentication attempt fails.



- 25. Return to **FGT-EDGE** and click **User & Authentication > Guest Management**.
- 26. The account **guest@acmecorp.net** is **Expired**.

| User ID ⓘ         | Expires ⓘ |
|-------------------|-----------|
| guest@example.com | Expired   |

**Index:** 8.0

**Use Case:** Setting up a Fortinet Security Fabric

**Objective Title:** New Objective

**Points:** 0

----- **Objective Section** -----

**Objective Text:**

## Setting up a Fortinet Security Fabric

In this lab, you configure FGT-EDGE as the root FortiGate in a Fortinet Security Fabric that also includes FGT-ISFW, FGT-DC, and a FortiAnalyzer.

The steps you complete to accomplish this lab are:

1. Create a Fabric connector to FortiAnalyzer
2. Create a Security Fabric group and authorize FGT-DC
3. Pre-authorize FGT-ISFW and add it to the Security Fabric
4. Authorize the FortiGate devices on FortiAnalyzer

Time to complete: 10 minutes

**Index:** 8.0 (a)

**Use Case:** Setting up a Fortinet Security Fabric

**Objective Title:** Create a Fabric Connector to FortiAnalyzer

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

## Create a Fabric Connector to FortiAnalyzer

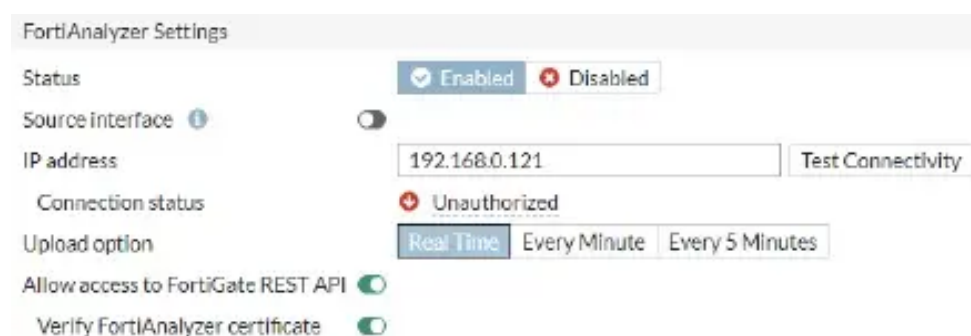
### Background

In the exercise, you create a Fabric connector on FGT-EDGE for the FortiAnalyzer.

**Note:** You can also add FortiAnalyzer as part of the process of creating a Security Fabric group, which you will do in the next exercise.

### Tasks

1. Return to the **FGT-EDGE**.
2. Click **Security Fabric > Fabric Connectors**. Under **Core Network Security**, select the connector **FortiAnalyzer Logging** and click **Edit**.
3. Set **Status** to **Enabled**.
4. Set **IP address** to 192.168.0.121, the IP address of the FortiAnalyzer.
5. Click **Test Connectivity**. The **Connection status** is currently **Unauthorized**.
6. Set **Upload option** to **Real Time**.
7. Leave the other settings at the default.



The screenshot shows the 'FortiAnalyzer Settings' configuration page. The 'Status' is set to 'Enabled' (checked). The 'Source interface' is set to 'any'. The 'IP address' is set to '192.168.0.121'. The 'Connection status' is 'Unauthorized'. The 'Upload option' is set to 'Real Time'. The 'Allow access to FortiGate REST API' and 'Verify FortiAnalyzer certificate' are both checked.

| Setting                            | Value         |
|------------------------------------|---------------|
| Status                             | Enabled       |
| Source interface                   | any           |
| IP address                         | 192.168.0.121 |
| Connection status                  | Unauthorized  |
| Upload option                      | Real Time     |
| Allow access to FortiGate REST API | Checked       |
| Verify FortiAnalyzer certificate   | Checked       |

8. Click **OK**.

9. A message appears about the FortiAnalyzer serial number. The correct number is **FAZ-VMTM19004582**. If this matches the number that appears, click **Accept**.
10. The connector is now blue. The red downward arrow shows that it isn't connected, because FGT-EDGE isn't authorized on the FortiAnalyzer. You will do this in a later step.



Index: 8.0 (b)

Use Case: Setting up a Fortinet Security Fabric

Objective Title: Create a Security Fabric Group and Authorize FGT-DC

Points: 10

----- Objective Section -----

Objective Text:

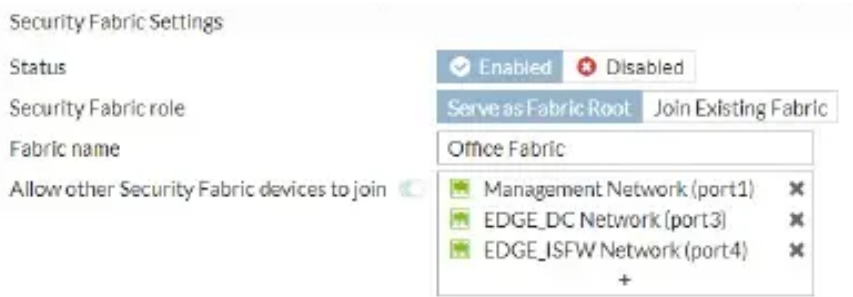
## Create a Security Fabric Group and Authorize FGT-DC

### Background

In this exercise, you configure FGT-EDGE as the root FortiGate in a Security Fabric group that includes it and the FortiAnalyzer. You also authorize FGT-DC as a member of the Security Fabric group. For this exercise, FGT-DC is pre-configured as a downstream Security Fabric device.

### Tasks

1. Click **Security Fabric > Fabric Connectors**. Under **Core Network Security**, select **Security Fabric Setup** and then click **Edit**.
2. Under **Security Fabric Settings**, set **Status** to **Enabled**.
3. Set **Security Fabric role** to **Serve as Fabric Root**.
4. Set **Fabric name** to `Office Fabric`.
5. **Allow other Security Fabric devices to join** shows the interfaces that allow access using the Security Fabric Connection protocol. This includes the two interfaces connecting to FGT-ISFW and FGT-DC, as well as the interface Management Network, which connects to network containing the FortiAnalyzer.

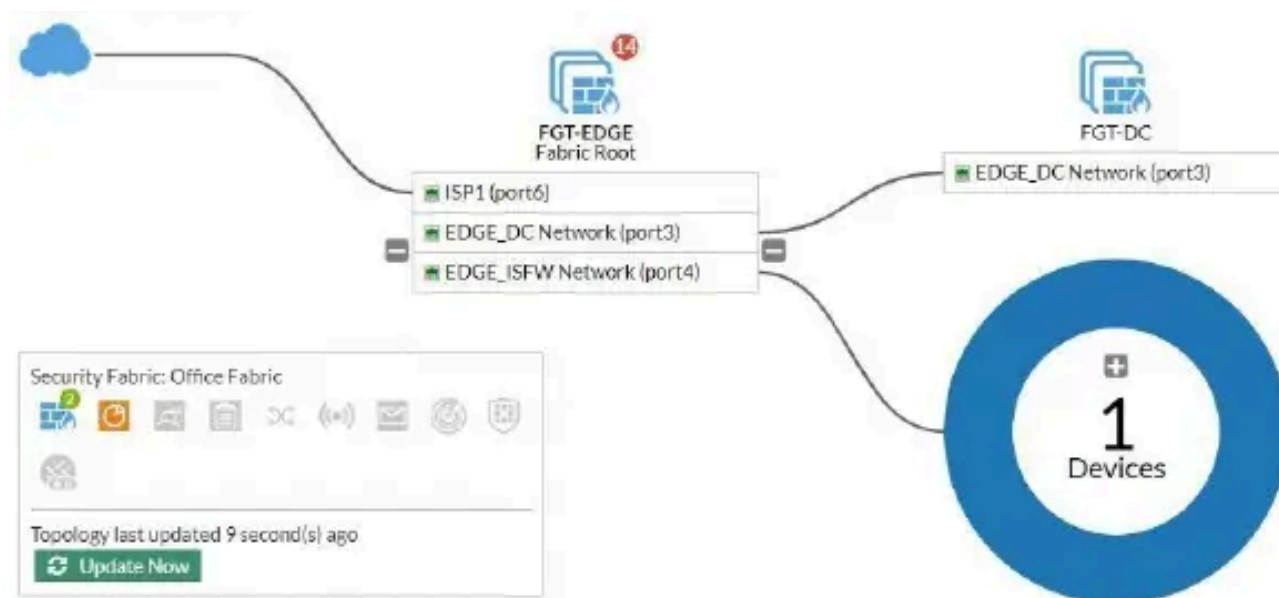


6. Click **OK**.
7. The connector is now red.

8. On the right side of the screen, under **Topology**, is a message asking you to authorize a device listed as FGVM01TM19002140. This is the serial number of FGT-DC. Click the device and select **Authorize** to add FGT-DC to the Security Fabric.



9. Click **Security Fabric > Logical Topology**. The topology displays the Security Fabric, including both FortiGate devices and FortiAnalyzer (the yellow icon in the box in the lower left corner represents the FortiAnalyzer).



**Index:** 8.0 (c)

**Use Case:** Setting up a Fortinet Security Fabric

**Objective Title:** Pre-Authorize FGT-ISFW and Add it to the Security Fabric

**Points:** 10

----- **Objective Section** -----

**Objective Text:**

## Pre-Authorize FGT-ISFW and Add it to the Security Fabric

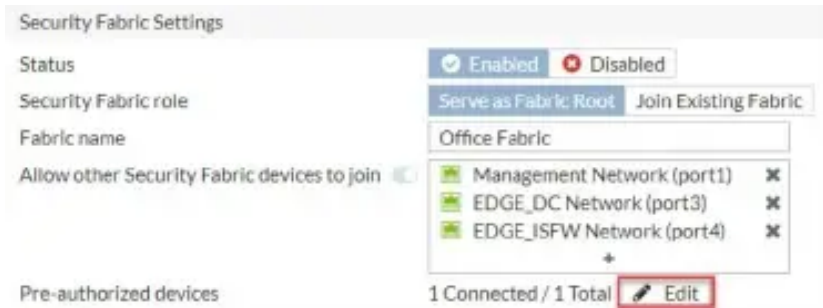
### Background

In this exercise, you pre-authorize FGT-ISFW as a Security Fabric device and then create a Security Fabric connector on FGT-ISFW.

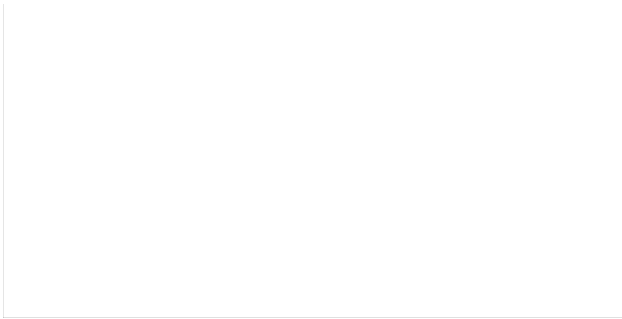
Pre-authorization simplifies the process of adding new FortiGate devices to a Security Fabric.

### Goal or Tasks

- 1. On the **FGT-EDGE**.
- 2. Click **Security Fabric > Fabric Connectors**, select **Security Fabric Setup**, and click **Edit**.
- 3. Beside Pre-authorized devices, click **Edit**.



- 4. In the right-hand window, FGT-DC is listed as an authorized device. Below FGT-DC, click + to add FGT-ISFW.
- 5. Set **Serial Number** to FGVM01TM19002141 (the serial number of FGT-ISFW) and leave **Action** set to **Accept**.



6. Click **OK**.
7. Click **OK** to save the Fabric connector.
8. Return to the Lab Activity Tab. Click **FGT-ISFW** in the sidebar menu under the Core group, and then click on the **HTTPS** option to access the **FGT-ISFW** device. Log in using username `admin` and password `Fortinet1!`
9. Click **Security Fabric > Fabric Connectors**. Select **Security Fabric Setup** and then click **Edit**.
10. Under **Security Fabric Settings**, set **Status** to **Enabled**.
11. Set **Security Fabric role** to **Join Existing Fabric**.
12. Set **Upstream FortiGate IP** to `10.10.30.14`, the IP address of port 4 on FGT-EDGE, which connects to FGT-ISFW.
13. Set **SAML Single Sign-On** to **Manual**.

Security Fabric Settings

Status: ☒ Enabled ☐ Disabled

Security Fabric role: ☐ Serve as Fabric Root ☒ Join Existing Fabric

Upstream FortiGate IP: 10.10.30.14

Allow other Security Fabric devices to join: ☒

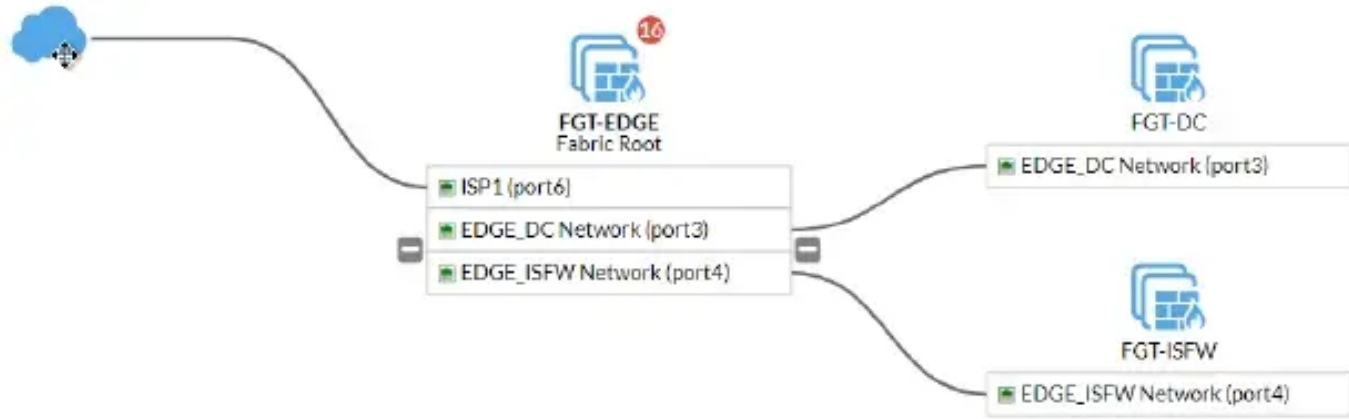
SAML Single Sign-On: ☒ Auto ☒ Manual ☐ Advanced Options

Mode: Disabled

Management IP/FQDN: ☒ Use WAN IP ☐ Specify

Management port: ☒ Use Admin Port ☐ Specify

14. Click **OK**.
15. Return to the **FGT-EDGE**.
16. Click **Security Fabric > Logical Topology**. The topology displays all three FortiGate devices.



**Index:** 8.0 (d)  
**Use Case:** Setting up a Fortinet Security Fabric  
**Objective Title:** Authorize the FortiGate Devices on FortiAnalyzer  
**Points:** 10

----- **Objective Section** -----  
**Objective Text:**

## Authorize the FortiGate Devices on FortiAnalyzer

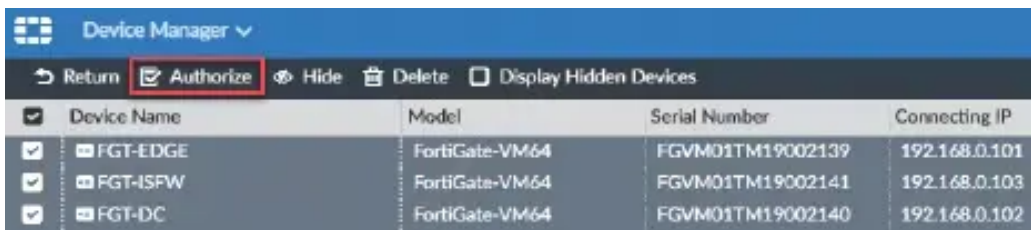
### Background

In the exercise, you authorize the FortiGate devices on the FortiAnalyzer so that the FortiAnalyzer will accept logs sent from these devices.

### Tasks

For this objective, we will be working on the **FortiAnalyzer**.

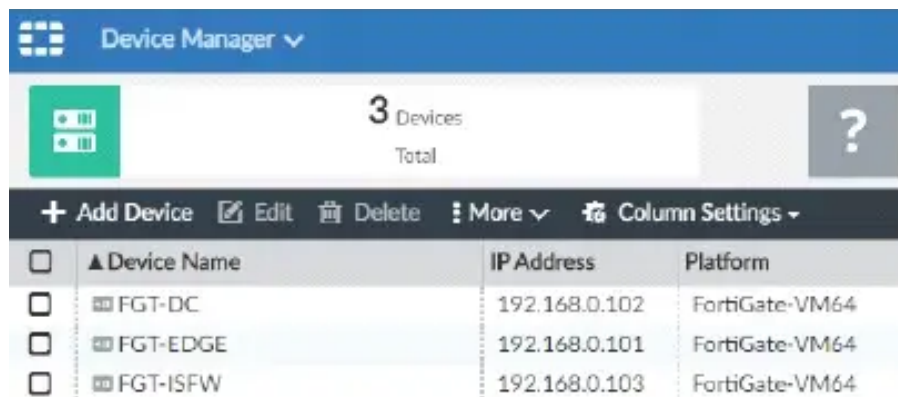
1. Return to the Lab Activity Tab. Click **FortiAnalyzer** in the sidebar menu under the Data Center group, and then click on the **HTTPS** option. Log in with the username `admin` and the password `Fortinet1!`
2. In **Select an ADOM**, click **root (3) Fabric**.
3. Click **Device Manager** and click **3 Devices Unauthorized**.
4. Select the devices and click **Authorize**.



The screenshot shows the 'Device Manager' window in FortiAnalyzer. The 'Authorize' button in the top toolbar is highlighted with a red box. Below the toolbar is a table with columns: Device Name, Model, Serial Number, and Connecting IP. Three devices are listed, all with checkboxes in the first column.

| <input checked="" type="checkbox"/> | Device Name | Model          | Serial Number    | Connecting IP |
|-------------------------------------|-------------|----------------|------------------|---------------|
| <input checked="" type="checkbox"/> | FGT-EDGE    | FortiGate-VM64 | FGVM01TM19002139 | 192.168.0.101 |
| <input checked="" type="checkbox"/> | FGT-ISFW    | FortiGate-VM64 | FGVM01TM19002141 | 192.168.0.103 |
| <input checked="" type="checkbox"/> | FGT-DC      | FortiGate-VM64 | FGVM01TM19002140 | 192.168.0.102 |

5. Set **Add the following device(s) to ADOM to root**.
6. Click **OK**.
7. When the authorization process is complete, click **Close**.
8. The devices are now on the list of authorized devices.



| Device Manager                                                                                                                |             |                    |                |
|-------------------------------------------------------------------------------------------------------------------------------|-------------|--------------------|----------------|
|                                                                                                                               |             | 3 Devices<br>Total |                |
| <a href="#">+ Add Device</a> <a href="#">Edit</a> <a href="#">Delete</a> <a href="#">More</a> <a href="#">Column Settings</a> |             |                    |                |
| <input type="checkbox"/>                                                                                                      | Device Name | IP Address         | Platform       |
| <input type="checkbox"/>                                                                                                      | FGT-DC      | 192.168.0.102      | FortiGate-VM64 |
| <input type="checkbox"/>                                                                                                      | FGT-EDGE    | 192.168.0.101      | FortiGate-VM64 |
| <input type="checkbox"/>                                                                                                      | FGT-ISFW    | 192.168.0.103      | FortiGate-VM64 |

9. Return to the **FGT-EDGE**.

10. Click **Security Fabric > Fabric Connectors**. The FortiAnalyzer Logging connector has a green arrow, showing that it is connected.



| Core Network Security                                                                                                                 |                                                                                                                                                                                                                                    |
|---------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Security Fabric Setup</p> <p>Office Fabric</p> |  <p>FortiAnalyzer Logging</p> <p>192.168.0.121</p> <p></p> |

## Stop and Think

True or false: you can pre-authorize a FortiGate device on the FortiAnalyzer before you configure the FortiGate log settings?

----- Answer Section -----

**Answer:** radio

**Answer Text:**

## Answer

True. From **Device Manager** on the FortiAnalyzer, you can click **Add Device** and enter information about a FortiGate to pre-authorize it before configuring the FortiGate log settings.

**Answer Key:**

- ✓ 1. True
- ✗ 2. False

**Index:** 8.0 (e)

**Use Case:** Setting up a Fortinet Security Fabric

**Objective Title:** Learn More About the Fortinet Security Fabric

**Points:** 0

----- **Objective Section** -----

**Objective Text:**

## Learn More About the Fortinet Security Fabric

If you would like to learn more about the Fortinet Security Fabric, as your instructor about the *Creating a Comprehensive Fortinet Security Fabric* course offered by the Fast Track Program.

Today's new world of networking requires a new approach to security that can do the following: simply, yet intelligently, secure the entire infrastructure; deliver full visibility into every viable network segment and the devices and endpoints behind them; and seamlessly integrate with third-party solutions, enabling users to ubiquitously collect, share, and correlate threat intelligence.

In this workshop, participants learn about the Fortinet Security Fabric, the first ever architectural security approach designed to dynamically adapt to today's evolving IT infrastructure. This multi-layered approach provides broad, integrated, and automated protection against sophisticated threats.

Participants who attend this workshop will learn how to:

- Introduce the Fortinet Security Fabric and the main business drivers
- Detail specific components that make up the Security Fabric
- Build a comprehensive solution to prevent, detect and respond to security incidents using the broad, integrated, and automated approach

9.0

Conclusion

New Objective

0

Section -----



## **You have successfully completed the Getting Started with the FortiGate Firewall Hands-On Lab**

### **Thank You**

To get more information on this or other Fortinet solutions, please consider looking at the NSE Training from Fortinet: <https://training.fortinet.com/>.

If you would like to learn more about the FortiGate, ask your instructor about the following available workshops from the Fast Track Program:

- Creating a Comprehensive Fortinet Security Fabric
- Constructing a Security SD-WAN Architecture
- What's New in FortiOS?
- Fortifying the Enterprise Network (NFGW Solution)
- Fortinet Teleworker Solution Engineered for Remote and Security Productivity
- SD-Branch: Securing Your Ethernet Switching Infrastructure with FortiSwitch, FortiAP, and FortiLink
- Attack and Defense Methodologies

- Cybersecurity for Safe, Reliable, Secure Industrial Control Systems (ICS)
- Security, Visibility, and Control of Public Cloud Infrastructure and Workloads

